AGAPPE

YOUR BEST PARTNER IN DIAGNOSTICS

USER MANUAL



MISPA-U280

Automatic Urine Analysis System

Copyright and Statement

Thank you for becoming a distinguished customer of Agappe and congratulations on your purchase of Automatic Urine Analysis System which will bring you convenience and a new experience.

This manual has been prepared according to the applicable laws and regulations. This manual contains the information up to the time when it is printed. Agappe reserves the rights for revision and interpretation of this manual and may update the manual without prior notice. Pictures used in the manual are for reference only; if any picture is inconsistent with the actual product, the latter shall prevail.

All data and information in this manual are protected by the copyright law. Without the written consent of Agappe, it is forbidden to reproduce, copy or translate any content of this manual into any other languages in any form.

The operator shall carefully read this manual before use and strictly follow the operating instructions described in the manual during operation. AGAPPE will assume no liability for any error or damage of the instrument caused by failure to follow the instructions herein.

Limited Warranty:

User Manual for Automatic Urine Analysis System defines the respective rights and obligations of AGAPPE and user in terms of quality assurance liability and after-sales services related to the product and provides for the creation and termination of such rights and obligations.

For an instrument sold by AGAPPE or its authorized agency, AGAPPE will provide one-year warranty services for defects in workmanship and materials from the date when the instrument is installed. AGAPPE's inspection is not flawless.

AGAPPE's obligation under such warranty is limited to repair of the product, which does not cover any economic loss or consequential cost caused by downtime of the instrument, including but not limited to:

Freight (including customs fees);

Economic and time losses arising from failure to use the instrument;

Accommodation, meals or other traveling expenses;

Losses arising from inconvenience;

Other costs.

AGAPPE will not be responsible for any direct, indirect or consequential damage caused by the following circumstances; and in such cases, AGAPPE shall no longer be held liable for the safety, reliability and running condition of the instrument, and all rights under the warranty shall be deemed as having been unconditionally and permanently waived.

- Abnormal use of the instrument, or use of the unmaintained or damaged instrument;
- Use of any reagent or accessory not provided or accepted by AGAPPE;
- Damage of the instrument caused by mis operation or negligence due to failure to follow the *User Manual for Automatic Urine Analysis System*;
- Replacement of any part without AGAPPE's permission;
- Maintenance, repair or modification of the instrument by persons without AGAPPE's authorization;
- Disassembly, assembly stretching or re-commissioning of components without permission.

NOTE: AGAPPE does not provide any implied warranties of merchantability and fitness of the instrument for a particular purpose.

You are welcome to contact us when encountering any technical problem during use. AGAPPE's After-sales Service Department has opened a service hotline to provide users with technical support and solutions.

In case of any fault of the instrument, please keep relevant sample and immediately notify your local authorized agency of Agappe

Product Name: Automatic Urine Analysis System

Manufacturer: Agappe



In Vitro Diagnostic medical device

Foreword

I. Safety Precautions

This instrument is a valuable precision electronic medical analyzer. In order to avoid damage to the instrument and personal injury caused by electric shock, cut off the main power supply before moving the instrument.

Use a regulated power supply with a power of 600VA or above.

Before you connect a new hardware device to the instrument, disconnect all power cords; first connect the data cable of the device, and then connect the power cords.

If you use a plug-and-play device in this instrument, we remind you that such operation could interrupt the running of the instrument or result in crash, and we will not guarantee the safety of the instrument and the plug-and-play device you use.

Before installing or using the instrument, carefully read relevant information provided in this manual. Use of the instrument without following the instructions specified by AGAPPE could compromise the protections provided for the instrument.

Strong light, dust, high temperature, dampness, violent vibration and shaking will affect the precision and service life of the instrument. Therefore, care should be taken to avoid such conditions.

During routine maintenance of the instrument, note that biological contaminants may exist in the auto sample loading unit, sampling probe, internal pipeline, waste pipe and waste tank; when cleaning and replacing these parts, wear disposable gloves or other protection equipment.

II. About the User Manual

This manual provides the relevant data and information about the use of the instrument. This is the best information guide to use the instrument. Please read the manual carefully before use of the instrument. If you read this manual, you will understand the features and operating steps of the instrument. You can retrieve what you need by quick search in the contents.

All persons who use, maintain or move the instrument shall read this manual.

Conventions used in this manual.

⚠ WARNING

Information on how to avoid injury of the operator or other persons.

⚠ CAUTION

Information on how to avoid damage of the instrument and unreliable analysis results.

⚠ NOTE

Operation steps or content requiring special attention.

III. About Package

When you receive the packaging boxes shipped by AGAPPE or its agency, please open each packaging box together with engineering technicians of or authorized by the manufacturer to check whether the standard accessories are consistent with those shown in the *Packing List* supplied with the instrument and whether there is any serious mechanical damage.

IV. Finding More Information

You can get more information about AGAPPE's products as well as upgrade information related to this instrument by the following approaches.

1. Website:

You can get the latest information about the Automatic Urine Analysis System from our website: http://www.agappe.com/

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Chapter 1 Introduction of instrument

1.1 Overview of instrument

The Automatic Urine Analysis System (hereinafter referred to as the "Instrument") is used in conjunction with special-purpose reagents for analysis of physical and chemical indexes in human urine and for shooting and storing images of formed elements and manual counting of them. Physical and chemical index test items include WBC, nitrite, urobilinogen, bilirubin, vitamin C, protein, occult blood, potential of hydrogen, specific gravity, glucose, ketone body, microalbumin, creatinine and urinary calcium.

The instrument consists of a urine dry chemistry analysis unit and a urine formed element analysis unit, which can work together or separately with a single sample for urine dry chemistry and/or urine formed element analysis. Also, the instrument has the function of analyzing the specific gravity, turbidity and conductivity of urine and other indexes by physical methods. The dry chemistry unit, in conjunction with suitable Urine Reagent Strips (hereinafter referred to as "strips"), can complete up to 14 test items. The formed element analysis unit uses a multi-channel counting cell; each channel has the forced cutoff function to make the sample become still rapidly; through auxiliary measures, quick sedimentation can be realized. The instrument can shoot and store images of formed elements in the sample for automatic identification. Test results are printed via an external printer connected to the instrument.

1.2 Intended use

The Automatic Urine Analysis System is a device intended to detect the formed elements and chemical constituents of urine.

1.3 Main structure and composition

The instrument consists of a urine dry chemistry unit and a urine formed element analysis unit. These units can be further divided into circuit control module, optical test system, PC, data processing module, pipeline module, mechanical module and microscope module.

The mechanical module mainly consists of sample loading assembly, sample adding assembly, strip selection system, etc.; the data processing module mainly consists of PC, CMOS image processor and system software; the pipeline module mainly consists of aspiration probe, connecting pipe, pump and solenoid valve; the microscope module mainly consists of microscope body, high and low power objectives and optical slide assembly (OSA). The circuit control module is mainly used to automatically control the orderly operation of each mechanical module and works with the data processing module to fulfill comprehensive processing of various data. The system software provides user-friendly human-machine interaction interfaces and is also responsible for data analysis, display, storage and printing of captured images. The pipeline module mainly completes liquid injection and discharge throughout the test process. Function of the entire system can be realized only with the effective cooperation of the mechanical structure; therefore, the mechanical module is the basis of each automated processing.

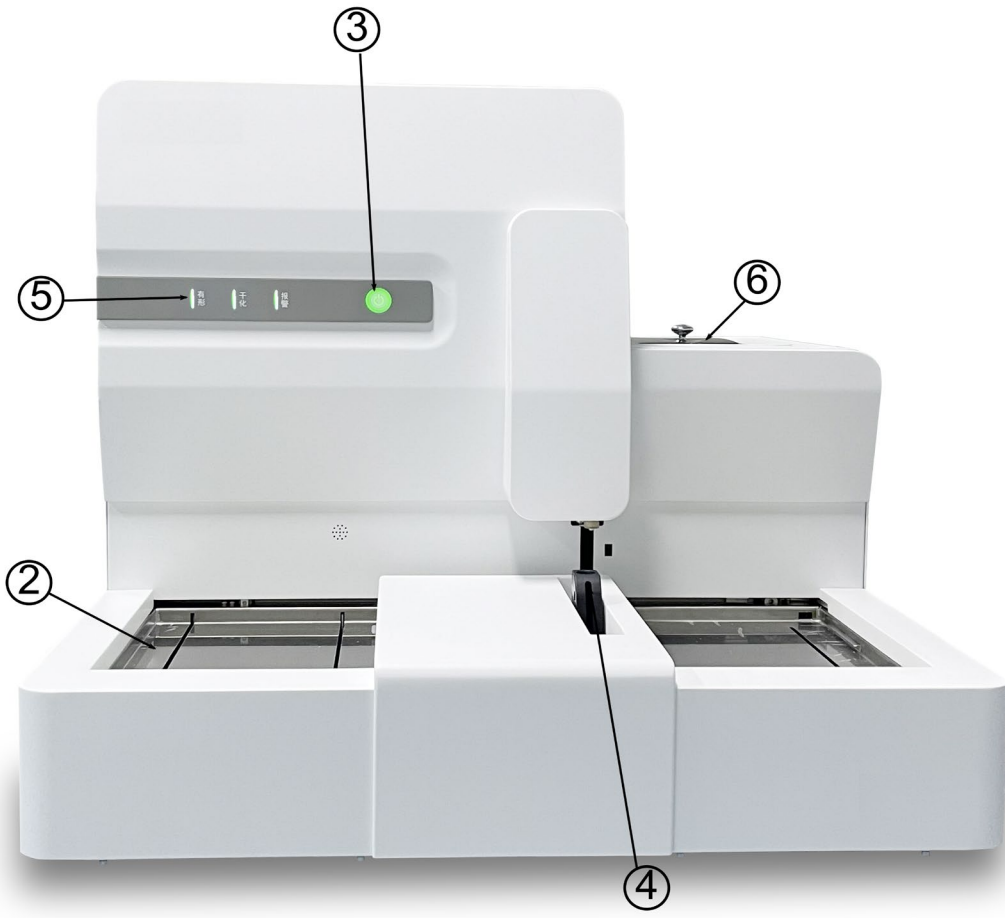


Fig. 1-1 Front view

① Test tube rack (including test tube rack A (see Fig. 1-2) and test tube rack B (see Fig. 1-3))

② Tray

③ Start button

④ STAT position

⑤ Indicator light

⑥ Test strip selection compartment

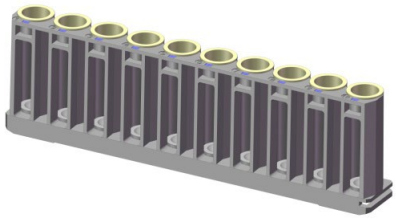


Fig. 1-2 Test tube rack A

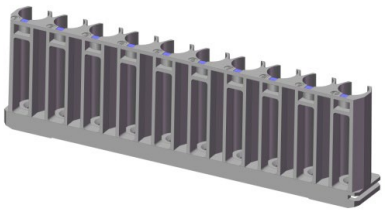


Fig. 1-3 Test tube rack B

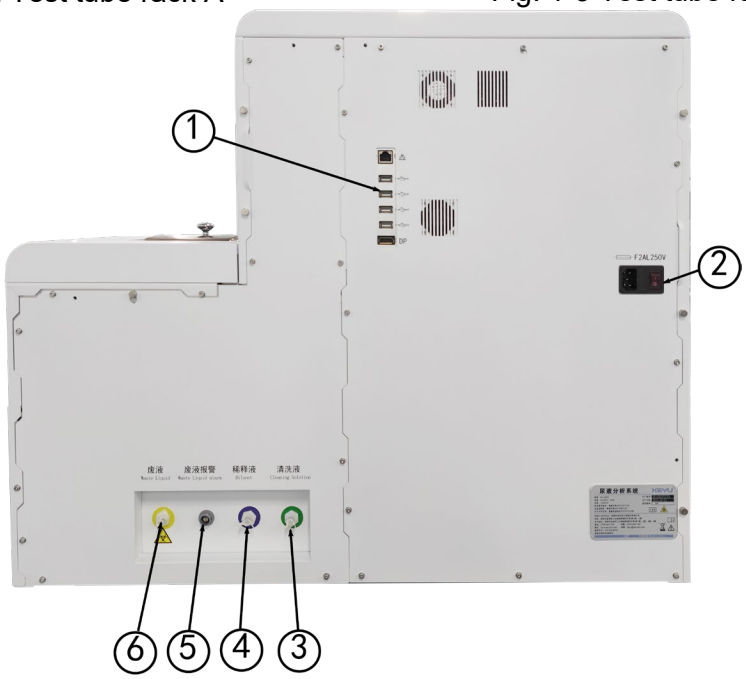


Fig. 1-4 Back view

① Hardware I/O port

② Power socket and switch

③ Cleaning solution port

④ Diluent port

⑤ Waste liquid alarm port

⑥ Waste liquid port

⚠ WARNING

- To avoid electric shock, do not open the cover of the instrument during work.
- To avoid electric shock, ensure that the instrument is well grounded.

1.3.1 Circuit control module

1) Control the opening and closing of solenoid valve for the on-off control the opening and closing of pipeline;
2) Control the opening and closing of pump to provide power for the pipeline;
3) Control the operation of sample loading stepper motor to provide power for the auto sample loading system;
4) Control the operation of microscope stepper motor to provide power for auto running of the microscope;
5) Control the light source of microscope;
6) Control the camera, automatically set relevant camera parameters and acquire images;
7) Control the operation of all lights and electric switches;
8) Communicate with the upper computer and transfer data.

1.3.2 Microscope module

The microscope module mainly consists of microscope body, high and low power objectives and optical slide assembly (OSA).

1) The OSA provides a carrier for the sample;

2) The microscope magnifies the formed elements in the sample.

1.3.3 Data processing module

It mainly consists of PC, CMOS image processor and system software.

1) Communicate with the electronic control unit to control the operation of the instrument;
2) Recognize the collected images and save the results;
3) Receive test results from other medical devices and communicate with the LIS;
4) Realize human-machine interaction.

1.3.4 Pipeline module

The pipeline module mainly consists of solenoid valve, pump and sampling probe, providing a channel and carrier for samples, reagents and liquid waste.

1.3.5 Mechanical module

It mainly consists of sample loading assembly, sample adding assembly, strip selection system, etc.

1.4 Reagent system

In order to maintain the best performance of the instrument, AGAPPE has developed a special reagent for KU-2800: Reagent for Automatic Urine Analysis System. All the reagents have been inspected and confirmed OK; The nominal test indexes of the instrument are all obtained under the condition of use of the Reagent Kit for Automatic Urine Analysis System. Use of reagents not supplied by AGAPPE will affect the performance of the instrument and lead to serious measurement errors, and could result in the occurrence of accidents. All reagents must be stored under room temperature to maintain the ideal chemical performance. The environment for reagent storage should not be too cold/hot and should not be exposed to direct sunlight; when the temperature is below 0℃, freezing of reagents could be caused easily, resulting in changes in the chemical performance and conductivity of reagents.

During use, in order to minimize reagent evaporation and reduce reagent contamination by external factors to the maximum extent possible, the reagent container shall be tightly closed with the container cover, and the pipe should enter the reagent through the container cover.

Note: The quality of reagents changes over time, so all reagents must be used within the shelf-life.

1.4.1 Diluent

The diluent is a type of stable isotonic solution used for:

- Maintaining the cell morphology during measurement;
- Providing photo background;
- Routine cleaning of the instrument (i.e., rinsing the inner and outer walls of sampling probe, the counting cell and the pipeline) to prevent cross-contamination.

1.4.2 Cleaning solution

- It is used for maintenance of the instrument in power-off state, which means the instrument will automatically fill all counting channels with cleaning solution before exiting the program every day. Upon startup next day, the instrument will automatically discharge cleaning solution out of the pipeline and wash the pipeline with diluent.

1.4.3 Maintenance solution

- The maintenance solution contains high-efficiency oxidant for deep cleaning of the instrument. It is used only when the counting cell or pipe system is seriously contaminated, for example, when there is a large number of spots in the microscopic image or plenty of viscous substance that cannot be washed off or removed by cleaning solution.

⚠ WARNING

- All reagents are used for in vitro tests; in case of contact with your eyes or skin,

rinse with plenty of running water.

⚠ CAUTION

- Please store or use the reagent kit according to relevant environmental requirements, and do not freeze or heat it; also keep the reagent kit away from any open power supply device, and use it within the shelf-life.

1.5 Reagent dosage

Diluent: Approx. 10 mL/sample

Cleaning solution: Approx.28 mL/shutdown

The reagent dosage changes with the use of the instrument.

1.6 Software operating environment

1.6.1 Hardware configuration

- Main frequency of CPU ≥1.7GHz.
- hard disk drive space ≥500G.
- RAM ≥2G.

1.6.2 Software environment

- System software: Operating system Windows 10 or a higher version
- Antivirus software: N/A
- Network condition: Network connection is required to communicate with the LIS or HIS
- Supported software: Opencv, Caffe

1.6.3 Security software update requirements

N/A.

1.7 Data ports

RJ45 network port, HL7 communication protocol.

The USB port can import and export data.

1.8 User access control

Account and password are required for login.

1.9 User permission control

There are two types of user account: Administrator and General User. The permissions of Administrator are different from those of General User.

Chapter 2 Operating principle and performance of instrument

2.1 Basic operating principle

The instrument applies the dry chemistry principle: The corresponding gradient value of each item is obtained by interpretation of the resulting color of urine test strip reaction.

The instrument adopts microscopy image-based urine sediment analysis: It uses the camera to shoot and store images of formed elements in urine and recognize the formed elements,and the result of formed elements are provided as non-auxiliary reference information for decision making,the doctor need to manually review and label the pictures and then issue the test results.

2.2 Operation process of instrument

The program controls the auto sample loading unit to convey the sample to the designated sampling point; after the sample is mixed evenly through the mixing pump and the mixing probe, the plunger pump, along with the sampling probe, dispenses liquid on the reaction block of urine analysis test strip for dry chemistry analysis and meanwhile aspirates a certain amount of sample to be tested to the OSA of each channel in sequence; the microscope magnifies the formed elements in the OSA for analysis.

A computer controlled camera synchronization device was used to collect multiple images of the samples distributed in the field of view for analysis and processing, and the samples with different properties were scanned by low power microscope and screened,and the small targets were identified by high power microscope.The different approach can improve the analysis speed and processing accuracy of the instrument. The image is stored, recognized, classified and statistics by software. Doctors need to manually review and label the images to form a standard urinalysis report,print various images. The

system can be additionally configured with a laser printer for printing analysis reports and images.

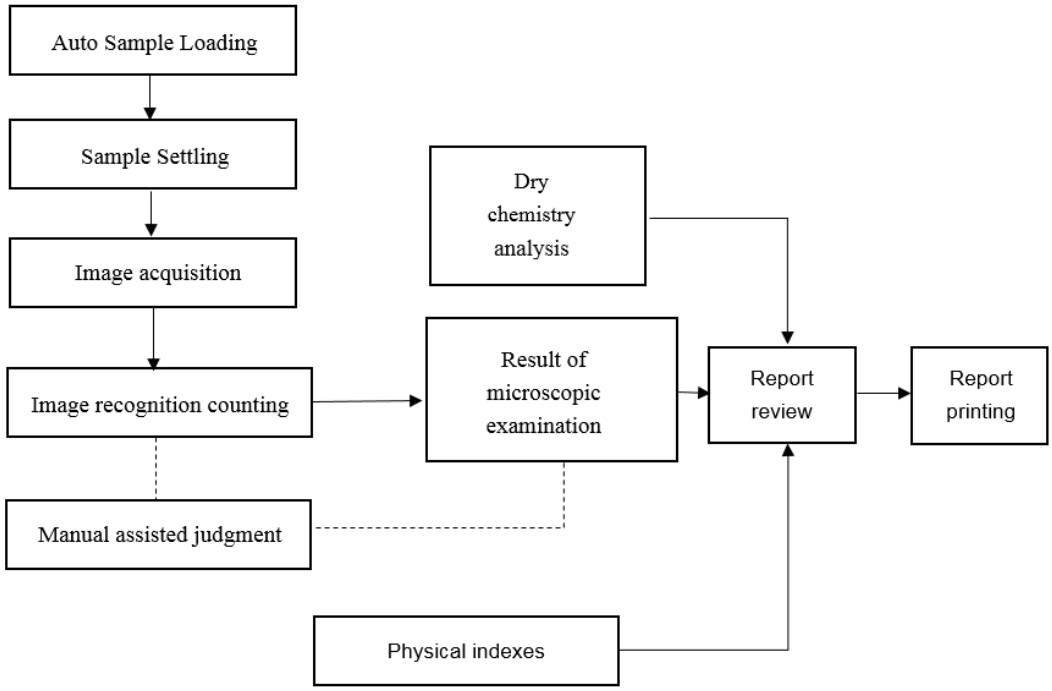


Fig. 2-1 Operation process

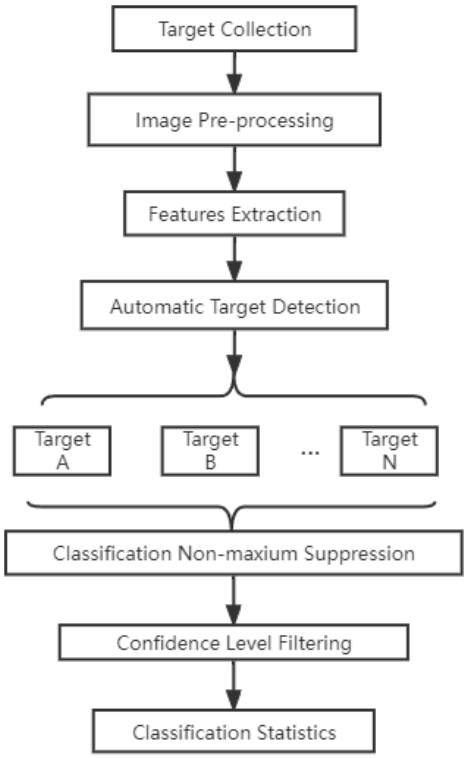


Fig. 2-2 Shooting process

2.3 Performance indicators

2.3.1 Appearance and structure

1) The appearance of the instrument should be tidy and clean and the surface coating and plating should be free of obvious peel-off, scratch and dirt.

2) The nameplate and marks should be clear.

3) Moving parts of the analyzer should be steady without getting stuck or having any "jump" phenomenon or obvious backlash.

2.3.2 Functions

1) Auto sample loading, auto strip selection, auto strip delivering, waste strip collection, auto bar code scanning (optional), auto mixing, quantitative sample aspiration/dispensing, auto washing, and auto interpretation.

2) Automatically control the microscope to achieve auto switching between high and low power images, auto return, auto focusing, auto tracking, auto adjustment of microscope brightness and auto positioning.

3) Define the review.

4) Input, edit, query and archive patient information.

5) It has clinical reference information functions such as image processing, initial automatic recognition and classification of formed elements, counting and basic statistics.

6) Edit, archive, query, import or export, review or reclassify and save results, and automatically generate formatted reports.

7) Generate routine urine analysis reports containing the results of dry chemistry analysis and formed element analysis and the test results of specific gravity, conductivity and turbidity of urine.

8) All results are provided with international units. The formed element analysis unit can automatically convert data input by manual identification to quantitative counting results expressed in the unit of "/μL".

9) Edit, store, query and browse reports.

10) Connect the LIS or HIS.

11) Automatically interpret the captured images of the reaction results of each dry chemistry item.

12) Store and memorize test data (optional) after power outage of the main unit.

13) Perform power-on self-test, and identify and report errors; have the output port; store test data; provide the QC calibration function.

14) The counting cell used for formed element analysis has 2 or more channels.

15) Provide the STAT function so that STAT samples can be tested immediately after arrival.

16) Realize auto batch test, up to 100 samples can be tested in each batch.

17) For dry chemistry analysis, 12 or 14 parameters of urine reagent strips are available.

2.3.3 Performance indicators of formed element analysis

2.3.3.1 Detection limit of formed element analysis

The analyzer should be capable of detecting red blood cells and white blood cells with the concentration being 10/μL

2.3.3.2 Repeatability of formed element analysis

The coefficient of variation (CV, %) of counting results from the analyzer should meet the requirements in Table 1.

Table 1 Coefficient of variation

Name of formed element	Concentration / (μL)	Coefficient of variation / (%)
Cell	50	≤25
	200	≤15

2.3.3.3 Recognition rate of formed element

1) Coincidence rate between individual results and microscope test results

The analyzer can at least automatically recognize red blood cell, white blood cell and urinary cylinder, and the coincidence rate between the individual results and the microscope test results should meet the requirements in Table 2.

Table 2

Formed Elements Name	Coincidence Rate/%
RBC	≥70
WBC	≥80
Urinary Cylinder	≥50

2) False negative rate

The false negative rate of analyzer's test result should less than 3%.

2.3.3.4 Stability of formed elements analysis

Within 8h after startup of the analyzer, the coefficient of variation (CV, %) of cell counting results should not be greater than 15%.

2.3.3.5 Carryover rate of formed element analysis

The carryover rate of the analyzer against cells should not be greater than 0.05%.

⚠Warning

The result of detection limit, repeatability, stability and carryover rate should be manually reviewed and confirmed.

2.3.4 Performance indicators of dry chemistry analysis

2.3.4.1 Repeatability of dry chemistry analysis

The coefficient of variation (CV, %) of reflectivity test results from the analyzer should not be greater than 1.0%.

2.3.4.2 Accuracy with special-purpose urine reagent strips

The difference between the test result and the corresponding nominal label-value of reference solution should not exceed one order of magnitude in the same direction, and there should be no difference in the reverse direction. Positive reference solution should not have negative result, and negative reference solution should not have positive result.

2.3.4.3 Stability of dry chemistry analysis

Within 8h after startup of the analyzer, the coefficient of variation (CV, %) of reflectivity test results should not be greater than 1.0%.

2.3.4.4 Carryover rate of dry chemistry analysis

Test the positive sample with the highest concentration result for test items (except specific gravity and pH), and then test the negative sample. The negative sample should not have positive result.

2.3.5 Test speed

1) Test time for single sample

The shortest test time of the analyzer for single sample does not exceed 100s.

2) Number of samples continuously tested per hour

The combined maximum testing speed for dry chemistry and formed element analysis is equal to or greater than 80 samples per hour.

2.3.6 Resolution

The resolution of images taken during microscopic examination is not less than 1600×1200.

2.3.7 Pipeline requirements

The pipeline system should not leak during normal working.

2.3.8 Environmental test requirements

a) Climatic environmental test should conform to the requirements for climatic environmental test Group II in Subclause 3.1 and the test requirements in Table 3 of GB/T14710-2009 as well as the temperature and humidity requirements for the test items.

b) Mechanical environmental test should conform to the requirements for mechanical environmental test Group II in Subclause 3.2 of GB/T14710-2009.

c) Transport test should conform to the requirements of Chapter 4 in GB/T14710-2009.

d) Power supply adaptability test should conform to the requirements of 5.1 in GB/T14710-2009.

2.3.9 EMC

Electromagnetic compatibility should conform to the requirements of GB/T 18268.1/ IEC 61326-1 and GB/T 18268.26/ IEC 61326-2-6.

2.3.10 Electrical safety

Electrical safety should conform to the requirements of the applicable clauses in IEC 61010-1:2010, A1:2016 , IEC 61010-2-101: 2018, GB 4793.1-2007, GB 4793.9-2013 and YY 0648-2008.

2.4 Technical parameters

1) The results of dry chemistry can be saved in the form of images.
2) The resolution of images of formed element analysis is not less than 1600×1200.
3) Sampling volume: Less than 3 mL.
4) Test speed:The combined maximum testing speed for dry chemistry and formed element analysis is equal to or greater than 80 samples per hour.
5) Maximum capacity of sample loading tray: 10 test tube racks, 100 samples.
6) Maximum number of test strips loaded single time: 200 strips.
7) Capacity of waste strip box: Containing up to 500 waste test strips.
8) Detectability of formed element: Detection rate in sample with concentration greater than 5/μL: ≥95％; cross-contamination rate: ≤0.5‰.
9) Report format: Summarize the tests (dry chemistry, physical analysis, formed element counting) in the report with both text and pictures.
10) Dimension of main unit: 778×713×650 mm
11) Weight of main unit: 80 kg (excluding the external monitor)
12) Optimum service life: The optimum service life of the instrument is 5 years; see the accompanying Certificate of Conformity for the manufacturing date of the instrument.

2.5 Power supply

Voltage: AC100-240 V;

Mains supply voltage fluctuations: 88-264 VAC

Frequency: 50/60 Hz;

Power consumption: ≤600 VA.

2.6 Storage and transport environment

Ambient temperature: -10℃~40℃;

Relative humidity: ≤93%RH.

2.7 Environmental conditions for operation

- Normal operating temperature: 5~40℃;
- Recommended ambient temperature: 15~30℃; (for the optimum operating temperature, see the instructions for use of test strips);
- Altitude: 2000 m
- Sound data: 60 dB
- Pollution degree of the intended environment: POLLUTION DEGREE 2
- Relative humidity: 20%~70%, non-condensing;
- Atmospheric pressure: 70kPa~106kPa;
- A good grounding environment should be available;
- The instrument should be placed on a fixed and stable workbench; the distance between the rear of main unit and the wall should not be less than 20cm;
- Keep away from sources of strong electromagnetic interference;
- Avoid direct sunlight;
- Keep away from sources of vibration and interference;
- The power socket should conform to the electrical code, and reliable grounding must be available.

Chapter 3 Installation of instrument

To achieve the optimum operation performance of the instrument and obtain satisfactory clinical effects, initial installation and commissioning of the instrument must be performed by engineers with AGAPPE's authorization; if the instrument is moved or used in another place, it must be installed and used according to the following installation procedure.

⚠ CAUTION

Installation of the instrument by any unauthorized or untrained person could result in damage of the instrument, and such damage is not covered by AGAPPE's free warranty. Without AGAPPE's authorization, no one shall install the instrument or use it in another place.

3.1 Unpacking and checking

Carefully take the instrument and its accessories out of the packaging box; properly keep the packaging materials for future transport or storage.

1) Check the instrument and accessories according to the accompanying Packing List;
2) Check whether there is any sign of water soaking;
3) Check whether there is any mechanical damage;
4) Check all exposed lead wires and inserted parts and accessories.

Please take the instrument and its accessories out of the packaging box together with the engineer authorized by the manufacturer or its agency.

In case of any problem, please contact AGAPPE's After-sales Service Department or agency immediately.

3.2 Installation requirements

1) When handling the instrument, put your hands in the gap at the bottom of the instrument and lift it up.

2) Selection and connection of power supply: Select an independent power supply as required; where necessary, an uninterruptible power supply (UPS) of ≥600VA should be equipped to ensure normal working.

3) Power-on check: Turn on all power switches to ensure normal data transfer and check whether each component of the instrument runs normally. Abnormalities should be handled by engineering technicians in the field.

4) When placing the instrument, do not block the power plug socket of the instrument to ensure that the plug can be disconnected in case of emergency.

5) Connection of communication cables: Monitor, keyboard, mouse and printer.

⚠ WARNING

- The instrument is not intended for home use.
- The instrument is not a therapeutic device.
- To move or transport the instrument after delivery, please contact our After-sales Service Department or agency.

⚠ NOTE

- The instrument should be protected from direct sunlight.
- The operating environment of the instrument should not be too cold/hot; if the air humidity is too high, water stain will exist on the surface of counting cell.
- Keep centrifuges and X-ray machines away from the instrument.
- Do not use mobile phones, radiophones or other devices generating strong radiation fields near the instrument, which may disturb normal working of the instrument.
- Do not place the instrument at a position where it is difficult to operate the disconnecting unit.

3.3 Check of power supply

Before installing the instrument, check whether the power supply on the installation site meets the following requirements for the instrument:

Voltage: AC100-240 V

Frequency: 50/60 Hz

⚠ WARNING

- The earthing rod on the rear panel of the instrument must be directly connected to the earthing conductor. User has the obligation to ensure the reliability of power supply protection ground.
- Before connecting the instrument to power supply, check to ensure that each electrical plug is connected correctly and securely.

⚠ CAUTION

- Frequent voltage fluctuation will result in degradation of the performance and reliability of the instrument; user shall solve this problem before use of the instrument, for example, by installing an additional AC voltage stabilizer.
- Frequent power interruption will result in serious degradation of the performance and reliability of the instrument; user shall solve this problem before use of the instrument, for example, by installing an uninterruptible power supply (UPS) (equipped by user).

3.4 Connection of reagent system

3.4.1 Connection of diluent pipe

Take the pipe with joint out of the reagent packaging box; connect the joint to the liquid connector marked with **diluent** on the rear panel of main unit; insert the other end of the

pipe into the diluent bottle, and tightly close the cap of diluent bottle. The diluent bottle must be placed on the same level as the instrument.

3.4.2 Connection of cleaning solution pipe

Take the cleaning solution pipe with joint out of the reagent packaging box; connect the joint to the liquid connector marked with **cleaning solution** on the rear panel of main unit; insert the other end of the pipe into the cleaning solution bottle, and tightly close the cap of cleaning solution bottle. The cleaning solution bottle must be placed on the same level as the instrument.

3.4.3 Connection of liquid waste pipe

Take the liquid waste pipe with joint out of the reagent packaging box; connect the joint to the liquid connector marked with **waste liquid** on the rear panel of main unit; clockwise tighten the bottle cap on the liquid waste pipe to the opening of liquid waste bottle. The position where the liquid waste bottle is placed should be at least 50cm below the plane where the instrument is placed.

3.4.4 Treatment of liquid waste

Liquid waste produced by the instrument is the mixture of urine sample and reagent; strong peculiar smell will be produced if liquid waste is placed for a long period of time, so liquid waste should be treated in time. Any organization using the instrument is suggested to strictly observe the national and local laws and regulations for treatment of medical wastes and strictly follow the methods specified in the medical waste disposal procedure of the organization.

⚠ WARNING 

- To prevent environmental pollution caused by liquid waste, it is forbidden to directly pour liquid waste to the drain; liquid waste must be biologically or chemically treated before discharge to the drain. Hospitals and laboratories have the obligation to observe the applicable regulations of the Environmental Protection Department of local government.

⚠ NOTE

- The special-purpose maintenance solution is used for deep cleaning and should not fed to the instrument through pipe.

⚠ CAUTION

- After installation of all pipes, leave the pipes in a natural and relaxed state without forced distortion, bending or turning.
- All pipe joints should be installed manually without using any tool.
- If damage or leakage of the reagent bottle, expiry of shelf-life or other abnormalities are found during installation, the reagent should not be used on the instrument; please contact your local office or supplier for replacement.

3.5 Connection of monitor, mouse and keyboard

1) Use the video cable to connect the monitor with the instrument;
2) Connect the power cord of the monitor;
3) Connect the mouse and keyboard to the instrument. If wireless keyboard and mouse are used, insert the wireless receiver into the USB port at the rear of the instrument, and install the driver.

3.6 Connection of power adapter

Insert one end of the power adapter into the power input socket of main unit, and the other end to a grounded three-phase power socket. Directly connect the earthing rod on the rear panel of main unit with the earthing conductor.

⚠ CAUTION

- The power adapter should be connected to a special socket.

3.7 Start of instrument

1) Turn on the UPS switch (if any);
2) Turn on the switch of special power strip connected with the main unit system;
3) Power on the peripheral device (e.g., printer, monitor);
4) Press the Start button on the left-side panel to start the instrument; it will enter the operating system.

3.8 Connection of external printer

1) Choose a proper position for placing the printer; it is suggested to place the printer on the right side of the instrument and keep a distance of at least 30cm from the instrument.
2) Install the accessories of the printer according to the Instruction Manual for the printer.
3) Connect the plug of printer cable to the corresponding socket on the rear panel of the instrument, and the other end to the printer.
4) Confirm that the power switch of printer is in OFF position; connect one end of the power cord to the printer, and the other end to the power outlet.
5) Install the communication cable; see the Instruction Manual for the printer for details.
6) Load printing paper according to the Instruction Manual for the printer.
7) Connect the power supply to initiate self-check of the printer.

⚠ NOTE

- On the first use, run the self-check program to test whether the printer is working normally; during subsequent use of the printer, the printing function of the printer can also be verified by running this program.

Chapter 4 Routine operations

4.1 Precautions

1) During operation, take care to prevent the sample from contacting human body, thus to prevent bacterial infection.
2) Ensure that the "Environmental Conditions for Operation" listed in 2.7 are met when the instrument is used.
3) If any abnormality is detected, cut off the power supply immediately to avoid damage of the instrument and short circuit.
4) In case of any fault of the instrument, please contact Agappe instead of attempting to repair the instrument since random repair could damage the instrument.
5) Do not place any container with liquid on the instrument to avoid liquid ingress and damage of the instrument.
6) After finishing the everyday tests, the instrument must receive routine maintenance to maintain the best performance.

⚠ NOTE

- Use of the instrument without following the methods specified by the manufacturer could compromise the protections provided for the instrument.

4.1.1 Sample

1) During test, the sample can be properly stirred but should not be centrifuged.
2) Use a clean and dry container to collect fresh urine and perform test as soon as possible. The urine sample should not stand for over 1h under room temperature; otherwise, the urine sample should be preserved in a refrigerator at 2℃~8℃ and tested within 2h. Do not add preservatives to the urine sample.
3) Do not disinfect or add detergent to the sample.
4) The sample should be protected from direct sunlight.
5) Injection or administration of vitamin C by the patient will result in low or false-negative test results of nitrite, bilirubin, glucose and occult blood.
6) Do not use bloody urine samples for test since contaminant will result in inaccuracy of test results. Through direct visual inspection, it can also be found that the color of bloody urine is inconsistent with the result tested by the instrument.
7) If the urine sample for test is collected after taking medicine, the test result could also be inaccurate.

4.1.2 Test strips

1) Please use special test strips provided by AGAPPE; use of other test strips will result in fault of the instrument.
2) Do not use any test strip beyond its shelf life. If the pad on the test strip is deformed, discolored or damaged, please do not use the test strip.
3) Take out test strips of the number required for test, and then immediately close the cap of test strip bottle to protect the unused test strips from dampness. If the test strips are exposed to air and not timely used for test, they will be affected with damp or contaminated by dust, resulting in inaccuracy of test results.
4) To avoid affecting the test results, do not touch the pad on the test strip.
5) Select a correct type of test strip before test; the type of test strip should be consistent with the display in the interface of the instrument.

4.2 Preparation before use

4.2.1 Check of wastes

1) Disposal of used test strips

Take out the waste strip box, and check whether there is any used test strip inside it; if any, remove the test strip.

2) Cleaning of liquid waste bottle

Check whether there is any liquid waste in the liquid waste bottle; if any, empty the liquid waste bottle.

4.2.2 Check of consumables

1) Cleaning solution and diluent

Make sure there is sufficient cleaning solution in the cleaning solution bottle and sufficient diluent in the diluent bottle; if insufficient, replenish the bottle with cleaning solution or diluent.

4.2.3 Preparation of test strips

1) Placement of test strips

Prepare test strips of the type required for test.

Check whether the type of test strips in the test strip delivery slot is correct; if not, empty the test strip delivery trough, and put the test strips in the corresponding test strip bottle; then tightly close the cap of test strip bottle. Up to 200 test strips can be placed in the test

strip delivery slot. When testing, load test strips of the number required for test from the test strip bottle.

⚠ NOTE

- Do not touch the pad on the test strip.
- Place test strips in the direction indicated on the test strip delivery slot.
- Tightly close the cap of test strip bottle to protect test strips from dampness.

2) Placement of desiccant

Take the desiccant out of the test strip bottle, and place it in the upper desiccant box ① or lower desiccant box ② of the test strip delivery slot; close the cover of test strip delivery slot (see Fig. 4-1).

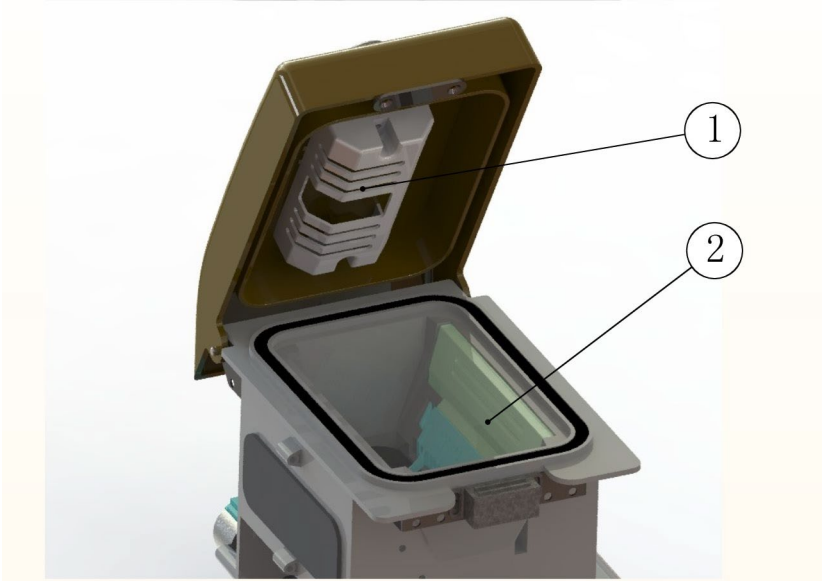


Fig. 4-1

4.2.4 Preparation of sample

1) Preparation of sample test tubes

Prepare clean sample test tubes.

If a bar code scanner is used, affix the bar code label about 10mm below the bottle opening.

2) Injection of sample

The sample volume for test is not less 3mL; if the sample volume is insufficient, inaccuracy of test results will be caused.

3) Placement of sample

Put the sample tube in the test tube rack; a single row of test tube rack can accommodate 10 sample tubes.

4) Placement of test tube rack

Place the test tube rack on the sample loading tray; at most 10 test tube racks can be placed in a single batch.

⚠ NOTE

- If the urine sample overflows, wipe the overflow before test; otherwise, crystallization could occur, further affecting movement of test tube rack.
- The test tube rack as a moving part should be placed in the sample loading tray according to the above arrow. In order to avoid personal injury, the instrument can start test only after your hands leave the test tube rack, and the test tube rack can be accessed only after the instrument stops.

4.3 General function interfaces

4.3.1 Startup/Shutdown

■ Startup

Turn on the power switch.

■ Shutdown

Click [Exit] at the upper right corner of the "Home" interface to exit the system; see Fig. 4-4.

4.3.2 Login interface

After startup, when system initialization (lasting for about 1~2min) is finished, the instrument will enter the Login interface.

After entering the user name and password, click <Login> to enter the System Initialization interface. You can click "Details" to view the content and progress of initialization or click <Access> to enter the "Home."

4.3.3 Home

When system initialization is finished, the "Report" interface will be opened automatically displaying the information of currently tested sample by default (see Fig. 4-4).



Fig. 4-4 Home

4.3.4 Test interface

Routine operation process

1) Turning on the instrument - Turn on the instrument according to the process described in 4.2 and 4.3.1.
2) Starting the operation software of the instrument - Double click the <KU-2800> icon on the PC desktop.
3) Login interface - Enter <User Name> and <Password>; see Fig. 4-2.

4) Sample placement - Put the test tube containing the sample in the test tube rack. Then put the test tube rack in the sample loading tray.

5) Starting test - Click the <Test> interface; click <Start> at the upper left corner; the instrument will start test. See Fig. 4-5.

6) Entry of patient information - Enter information in the "Patient Information" bar in the <Test> interface; patient information can be entered at any time.

7) Report review - When the test is finished, the sample result can be reviewed at any time.

8) Report printing - When the sample result has been reviewed, the report can be printed at any time.

9) Turning off the instrument - Click <Exit>; the instrument will shut down automatically for maintenance and exit the software; then turn off the PC.

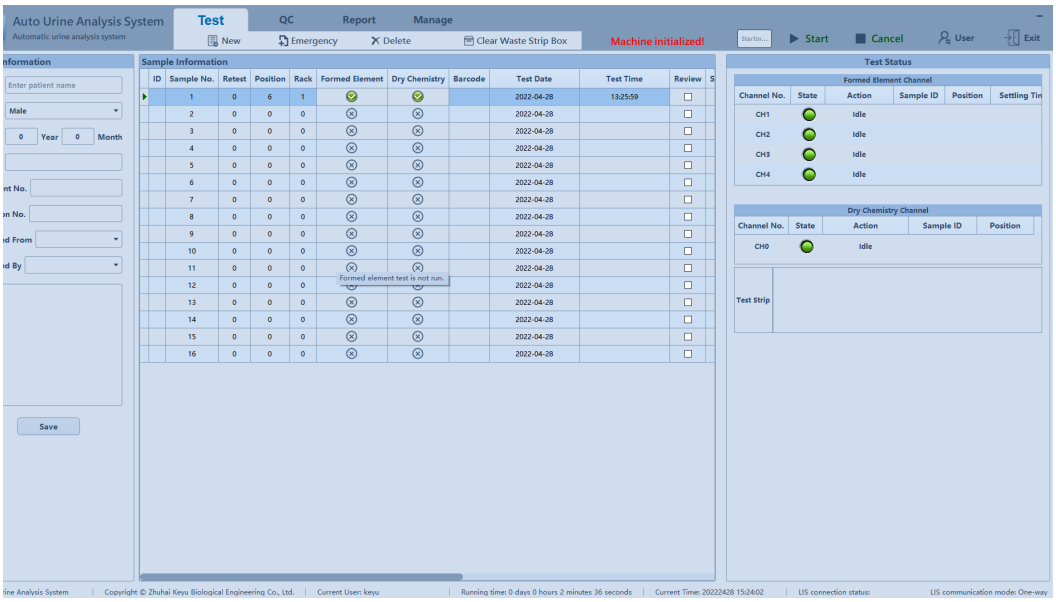


Fig. 4-5 Test interface

4.3.5 Sample info interface

1) Sample creation and deletion

Sample IDs are differentiated by date. Every day the regular sample IDs start from "1" by default, and emergency sample IDs start from "1000" by default. Also, you can create a specified sample ID: Enter the sample ID you want to create in the input box

(

) at the upper right corner of the interface; the system will create a usable sample ID according to this sample ID.

2) Manual creation of regular sample

Before sample test, you can create new regular sample information at any time. At the bottom of the <Test> interface, click <New> to quickly create a new regular sample. (See Fig. 4-6)

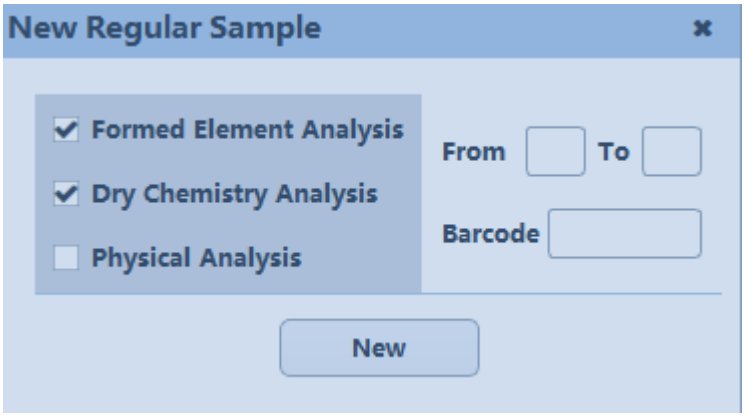


Fig. 4-6 Creation of new sample

3) Auto creation of new sample

During test, the instrument can add samples automatically; The system will apply auto numbering to sample IDs. If you enter a sample ID in <Starting Sample ID>, the sample ID auto numbering will start from the <Starting Sample ID>. <Starting Sample ID> is "1" by default; at this moment, the system will automatically retrieve and number sample IDs.

4) Scanning sample bar code

The instrument can scan the bar code of a new sample via the external bar code scanner; you can also enter the sample bar code after creation of a new sample.

5) Sample deletion

You can select the sample and click <Delete> to quickly delete the sample selected.

6) Entry and modification of sample information

In the <Sample Information> interface, select the sample ID. You can enter relevant information of the sample on the left side, and then click <Save>. See Fig. 4-7. During or

after test, you can edit sample information at any time. In the <Sample Information> interface, you can also view the test status and other basic information of the sample.



Fig. 4-7 Entry and modification of sample information

7) Modification of sample test items

For a sample waiting for the test, the test items can be selected by ticking in the drop-down box. (See Fig. 4-8)

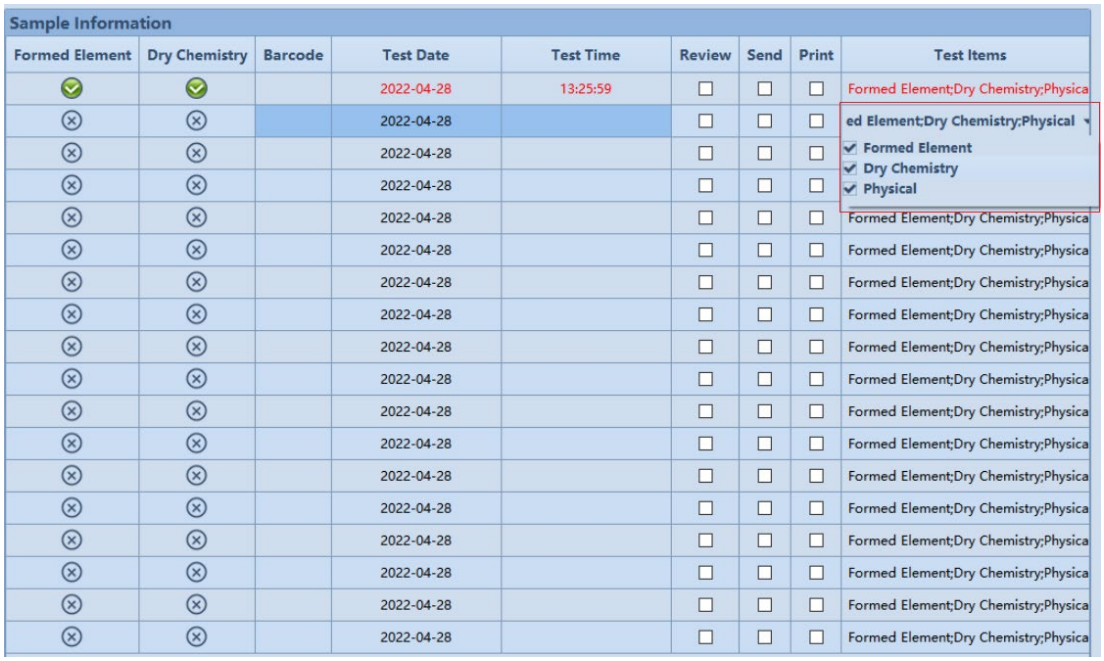


Fig. 4-8 Modification of sample test items

8) Modification of sample No.



Fig. 4-9 Modification of sample No.

No matter whether a sample has been tested or not, you can modify the sample No. by clicking <Modify Sample No.> in the context menu. (See Fig. 4-9 and 4-10)

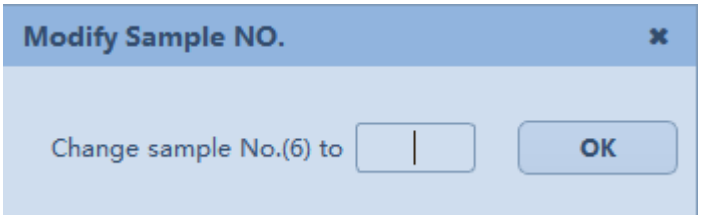


Fig. 4-10 Modification of sample ID

4.3.6 Report interface

4.3.6.1 Report interface

When sample test is finished, sample review can be carried out. After selecting the desired sample in the <Report> interface, the information bar on the left side will display the sample information and patient information, and the result bar on the right side will show the results of formed element analysis, dry chemistry and physical analysis of the sample.

The instrument takes pictures of the urine sample and uses AI intelligent software to identify the physical properties of the urine sample. It mainly refers to color and turbidity.

The color include unknown, colorless, light yellow, yellow, dark yellow, dark brown, light red, red, light green, green, and milky white.

The turbidity include clear, low turbidity, medium turbidity, high turbidity.

The instrument has individual detection channel, which can detect conductivity and specific gravity.

The conductivity uses resistivity method and the limit range is 2-655, conductivity has important clinical significance for patients with kidney disease.

The Specific gravity uses optical refraction method and the limit range is 1.000-1.040. Specific gravity has important clinical significance for patients with kidney disease.

Conductivity and specific gravity can be selected whether to use this function or not, and each laboratory is recommended to choose according to their needs.

There is no correlation between the two results of physical specific gravity and dry chemistry specific gravity,we suggest to refer to the physical specific gravity results.

Education with clinicians is required and it is recommended to focus on the physical specific gravity results. See Fig. 4-11 and 4-12.

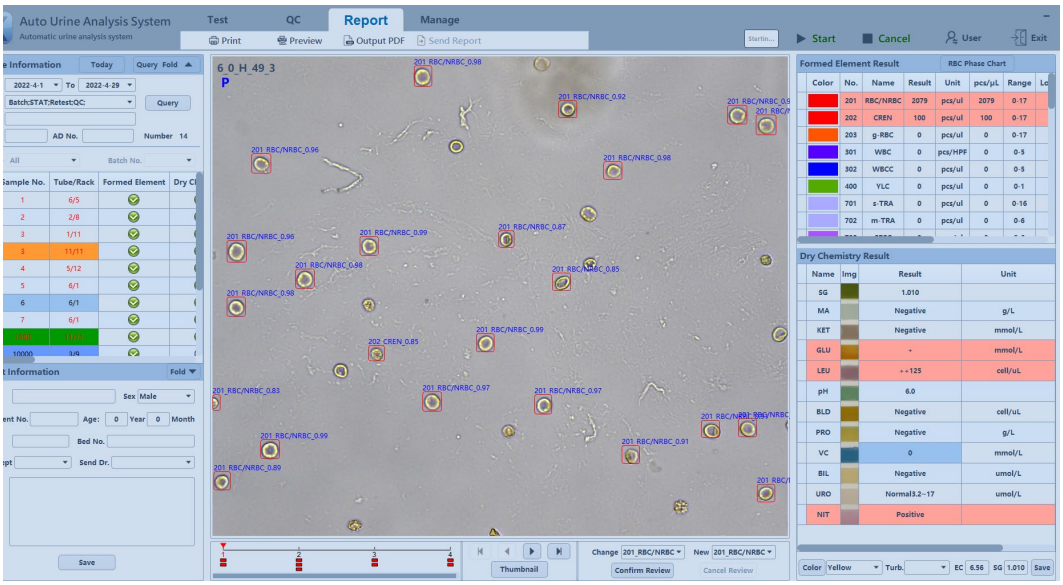


Fig. 4-11 Report interface



Fig. 4-12 Report interface

Our urine instrument has powerful photo and AI recognition functions, the instrument can take multi-layer pictures of relevant formed elements in urine and compare the pictures with the standard data in the database, then, the analyzer will mark and count the formed

elements in the pictures, finally, we can get the results through the unique algorithm of the urine instrument.

Nowadays, our urine instruments are able to recognize the following relevant formed elements:

1. Normal red blood cell
2. Crenated erythrocyte
3. Ghost RBC
4. Microcyte
5. Acanthocyte
6. Dysmorphic red blood cell
7. White blood cell
8. Pus cell
9. Yeast-like colony
10. Deep transitional epithelium
11. Squamous epithelial cell
12. Hyaline cast
13. Granular cast
14. Calcium oxalate crystals
15. Uric acid crystal
16. Phosphatic crystal
17. Mucus
18. Sperm
19. Bacterium

Up to now,our urine instruments still can't recognize these formed elements as follows:

1. Superficial transitional epithelium
2. Middle transitional epithelium

3. Transitional epithelium

4. Renal tubular epithelium

5. Red cell cast

6. White cell cast

7. Epithelial cast

8. Fatty cast

9. Waxy cast

10. Renal failure cast

11. Pathologic cast

12. Non-crystal urate

13. Leucine crystal

14. Cystine crystal

15. Bilirubin crystal

16. Cholesterol crystal

Reasons for urine instrument failing to identify or misidentify relevant formed elements: The instrument's database have no model similar to the formed elements in the picture, or the formed elements in the picture is extremely similar to the misidentified model in the database, and our instrument can not recognize all the formed elements correctly 100%.

Reporting principle: Due to the instrument identification failure or error, the laboratory physicians need to manually review all images one by one and make corrections accordingly by adding or removing the unrecognized and misrecognized elements.

The urine test is an extremely important part of the urine test, and many pathological diseases are judged on the basis of the urine test results, so it is important to ensure the accuracy of the test results, so we strongly request clinicians to manually review the test results to ensure the accuracy of the results.

4.3.6.2 Formed element review

There are two ways to review sample images: Thumbnail review(as shown in Fig. 4-13) and Classification Diagram(as shown in Fig. 4-14). When selecting samples in the <Sample Information> list, the images of samples will be displayed according to the

currently selected <Thumbnail> or <Classification Diagram> counterparts, and <Thumbnail> will be selected by default；

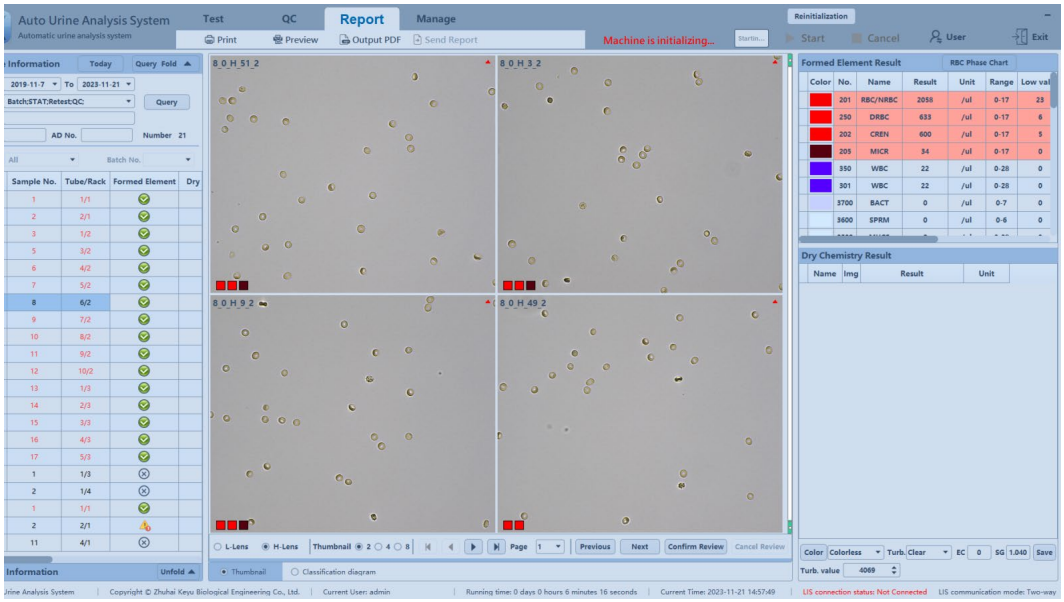


Fig. 4-13 Thumbnail Review



Fig. 4-14 Classification Diagram

In <Classification Diagram> the category displays the test items in which the number of formed elements in the current sample test is greater than zero; the currently selected <Category> item will be displayed according the settings preset in <Set Test Items> by low or high power lens ; as shown in Fig. 4-15.

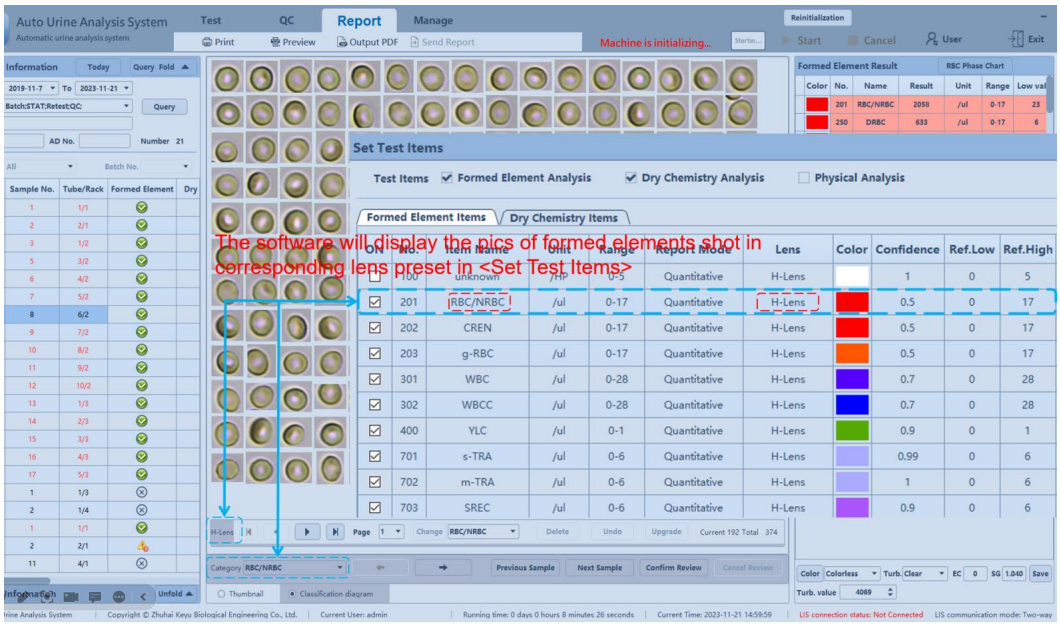


Fig. 4-15 Displayed according to the multiplier set for the project

Double-click on an image of formed element in the classification diagram to access the large image for review, and display the current location of the corresponding formed element and its location in the large image for the number of photos taken；as shown in Fig. 4-16.

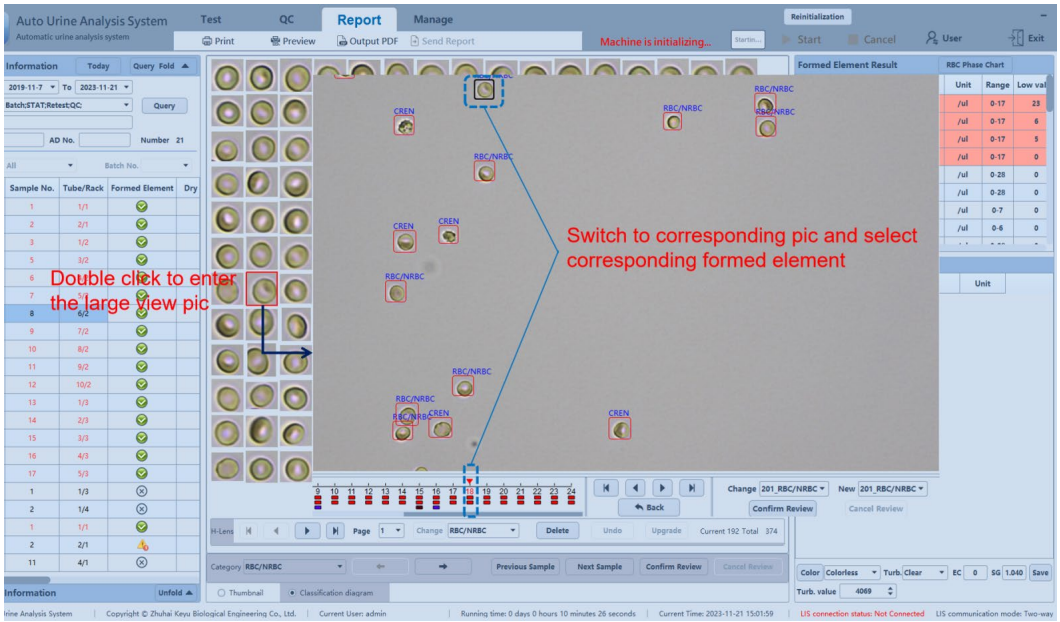
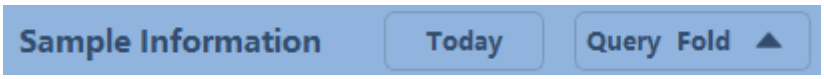


Fig. 4-16 Displayed according to the multiplier set for the project

The user can single-select, multi-select (press Ctrl + click), and mouse box-select to select the classification diagram for <Delete>, <Classification Diagram>, <Undo>, <Upgrade>, etc.; after executing the <Upgrade> operation, it will synchronize with the result of the corresponding item; when switching between items in the <Classification Diagram>, changing pages, or switching between the previous/next samples for auditing, etc., if there are formed elements in the current sample that are being operated by <Delete> or <Change>, it will automatically execute the <Uprade> operation, and synchronize with the result of the updated sample. as shown in Fig. 4-16.

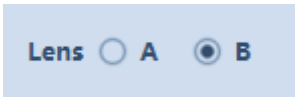
4.3.6.3 Buttons in the review report interface

1) 

Sample Information: Click <Sample> to show all information of samples tested on the current day; click <Query Fold/Unfold> to show/hide the query area.

2) 

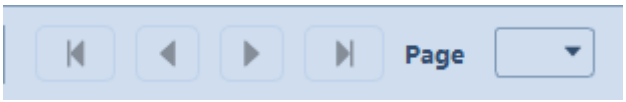
Patient Information: Click <Fold/Unfold> to show/hide the patient information queried.

3) 

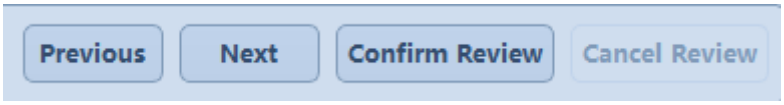
Switch between low magnification objective A and high objective objective B during review.

4) 

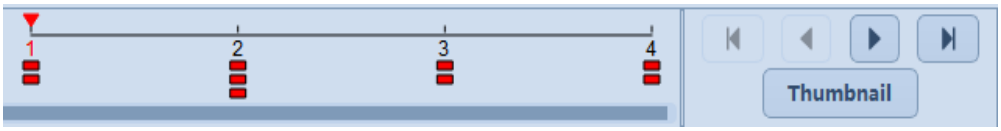
Show the number of thumbnails reviewed.

5) 

Page switching button: The specific current page can be selected in the drop-down box.

6) 

Click <Previous> or <Next> to switch the sample for browsing.

7) 

You can change views with the view switching slider and switching buttons. You can click the slider or buttons to locate to any group, and then realize layer switching via the mouse wheel; click <Thumbnail> to go back to the Thumbnail interface. Scrolling the mouse wheel in the image area can change different layers, outside the image area is to change different numbers of images.

8) 

Select a marked item in the big interface; the target category can be modified via the "Change" drop-down menu. You can also select an item in the big interface and add a new target category via the "New" drop-down menu. Click <Confirm Review> to confirm that the currently selected sample has been reviewed, and "Review Status" of sample information will be ticked ☑. Click <Cancel Review> to cancel the current reviewed status of the sample, and "Review Status" of sample information will be unticked ☐.

9) 

Move the mouse cursor to the <RBC Phase Chart> button; the red blood cell phase image will pop up automatically. (See Fig. 4-17)

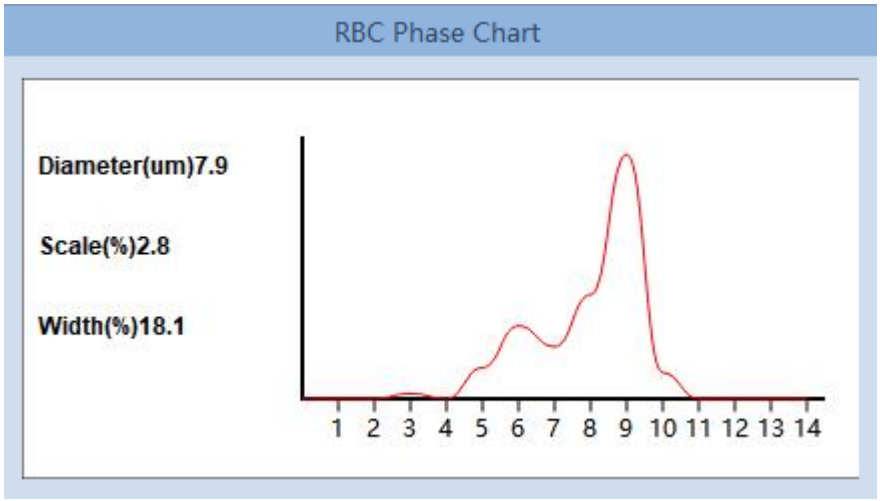


Fig. 4-17 Red blood cell phase image

10) 

Click <Save> to update the results of color, turbidity and specific gravity of the sample. Move the mouse cursor to the <Color> button; the sample color will pop up automatically, double click the picture can enlarge the picture. (See Fig. 4-18)

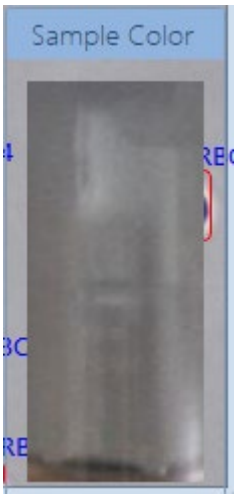
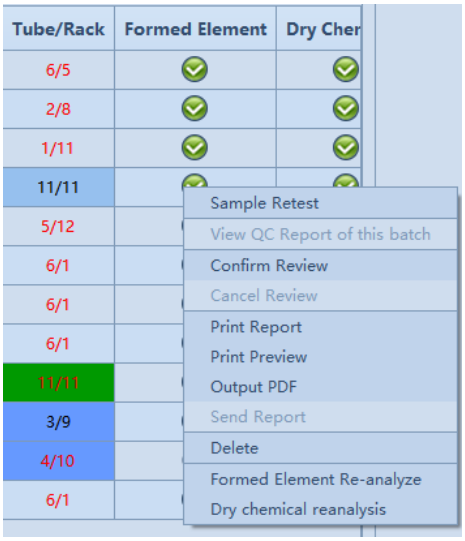


Fig. 4-18

11) 

Click <Print> to print the currently selected sample report; click <Preview> to preview the currently selected sample report; click <Output PDF> to output the sample report in PDF format; click <Send Report> to send the current sample information to the LIS.

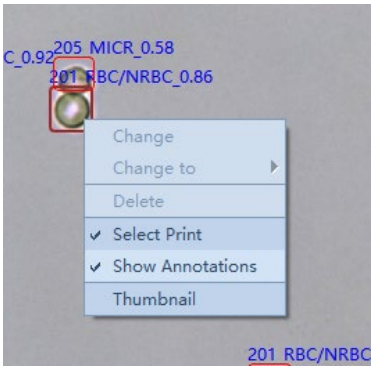
12)



Context menu in the Sample Info List: Click <Sample Retest>, the currently selected sample will be retested. Click <View QC Report of this batch> to view the QC Report of the currently selected QC sample.

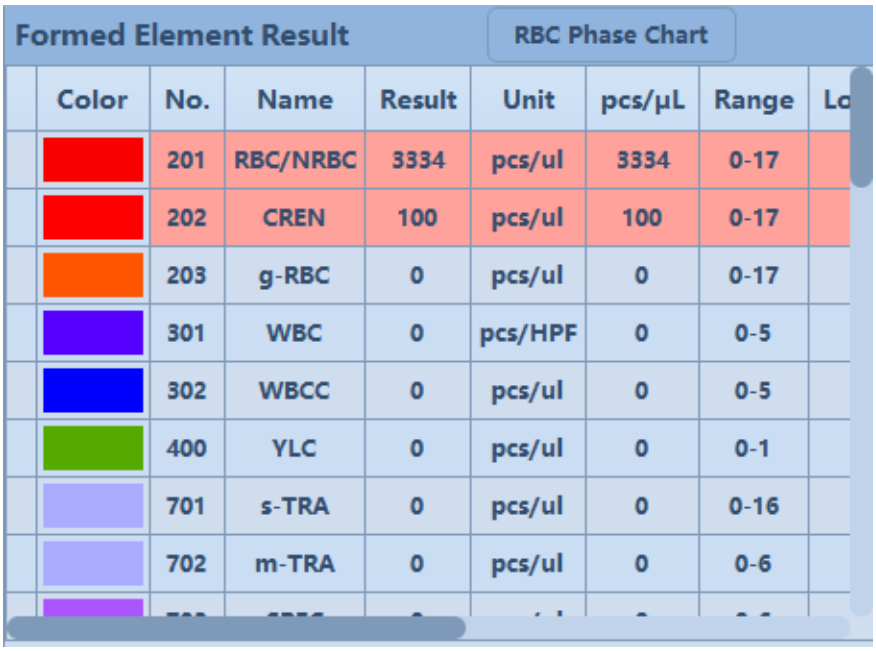
The functions of the <Confirm Review>, <Cancel Review>, <Print Report>, <Print Preview>, <Output PDF>, <Send Report>, <Delete>, <Formed Element Re-analyze> and other buttons are identical with those buttons with the same names in the interface. Click <Output PDF> to open the PDF file output for the current sample; click <Dry Chemistry Re-analyze> to re-analyze the test strip image of the current sample and update the dry chemistry result.

13)



Context menu in the big review interface: Click <Change> or <Change to> to change the currently selected mark to the item in the current drop-down options. Click <Select Print>; the selected image will show the letter "P". If the number of images selected exceeds the maximum allowed number, the images that are at the top of the list will be printed by default. Click <Show Annotations> to hide or show the annotations in the image; click <Thumbnail> to go back to the thumbnail of this sample.

14) You can directly modify the result in the Formed Element Result interface. (See Fig. 4-19)



Color	No.	Name	Result	Unit	pcs/μL	Range	Lo
	201	RBC/NRBC	3334	pcs/ul	3334	0-17	
	202	CREN	100	pcs/ul	100	0-17	
	203	g-RBC	0	pcs/ul	0	0-17	
	301	WBC	0	pcs/HPF	0	0-5	
	302	WBCC	0	pcs/ul	0	0-5	
	400	YLC	0	pcs/ul	0	0-1	
	701	s-TRA	0	pcs/ul	0	0-16	
	702	m-TRA	0	pcs/ul	0	0-6	

Fig. 4-19 Formed Element Result

15) You can directly change the result by double clicking the result bar and selecting from the drop-down options in the Dry Chemistry Result interface. (See Fig. 4-20)

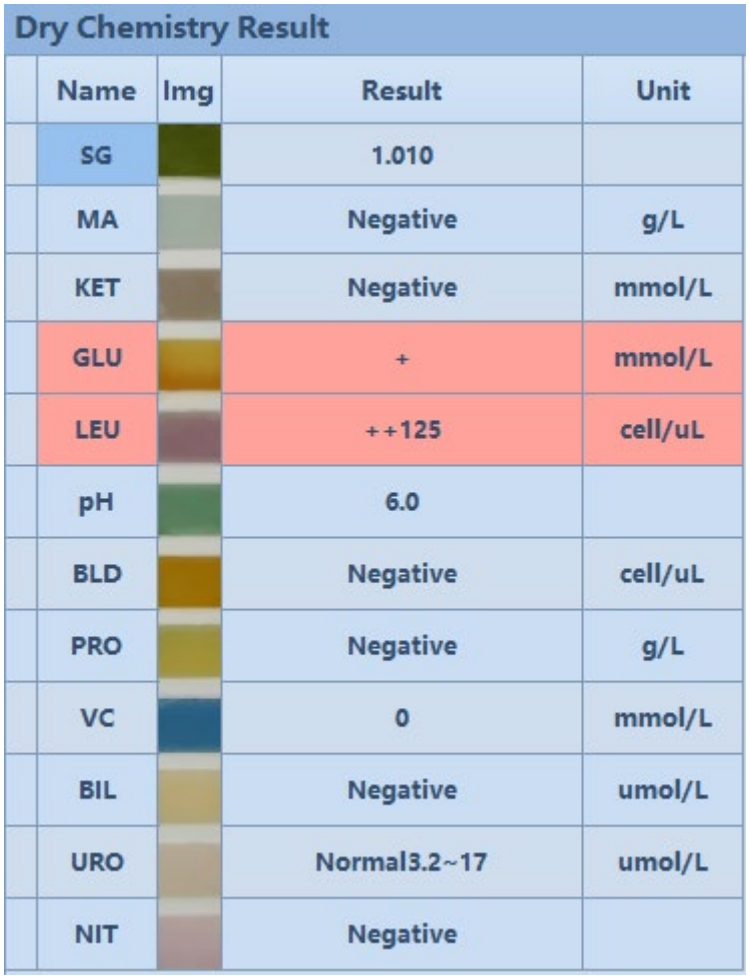


Fig. 4-20 Dry Chemistry Result

When the urine analyzer performs the dry chemical test, after the urine sample is dropped onto the urine test strip and reacts for a little time, the instrument will irradiate the test strip with the detection light, different test items on the urine test strip will show different color changes, different test items will absorb the irradiated monochromatic light, the instrument will compare the reflected light intensity of each test strip with the reflected light intensity of the blank test strip. A semi-quantitative result is obtained by calculation.

The 12-items and 14-items urine reagent strips are available.

The 12 & 14 urine reagent strips contain the following items:

Test items of 14 -items test strip: **Test items of 12 -items test strip:**

1. Nitrite 1. Nitrite

2. Urobilinogen	2. Urobilinogen
3. Bilirubin	3. Bilirubin
4. Vitamin C	4. Vitamin C
5. Protein	5. Protein
6. occult blood	6. occult blood
7. pH	7.pH
8.Leukocyte	8.Leukocyte
9.Glucose	9.Glucose
10.Ketone body	10.Ketone body
11.Microalbumin	11.Microalbumin
12.Specific gravity	12.Specific gravity
13.Calcium	
14.Creatinine	

Before the dry chemistry test, be sure to select the corresponding test strip type, otherwise the instrument will report an error and not get the dry chemistry results.

16）Alarm for Hematuria, Pyuria, Bacteriuria

Alarm for Hematuria

If sample test contains microscope items and dry chemistry, and the colour of urine is light red, or red, or dark brown, dry chemistry BLD 3+, RBC density high; the test status icon would turn red, and the system would alarm with “The sample concentration may be too high, recommended dilution retest”. when cursor staying on the item in sample information list. The system would also alarm with “The sample concentration may be too high, recommended dilution retest” when user click 【confirm review】.

If sample test only contains microscope items only, and the colour of urine is light red, or red, or dark brown, dry chemistry BLD 3+, the test status icon turn red, the system would alarm with “The sample concentration may be too high,

recommended dilution retest”when cursor staying on the item on sample information list. The system would also alarm with “The sample concentration may be too high, recommended dilution retest” when user click 【Confirm review】.

Alarm for Pyuria

If sample test contains microscope items and dry chemistry, and the colour of urine is white, or light green, or green, and dry chemistry BLD 3+, and cell density high; the test status icon would turn red, and the system would alarm with “The sample concentration may be too high, recommended dilution retest” when cursor staying on the item in sample information list. The system would also alarm with “The sample concentration may be too high, recommended dilution retest” when user click 【Confirm review】.

If sample test only contains microscope items only, and the colour of urine is white, or light green, or green, the test status icon would turn red, and system would alarm with “The sample concentration may be too high, recommended dilution retest” when cursor staying on the item in sample information list. The system would also alarm with “The sample concentration may be too high, recommended dilution retest” when user click 【Confirm review】.



Fig. 4-21 Alarm for hematuria, pyuria

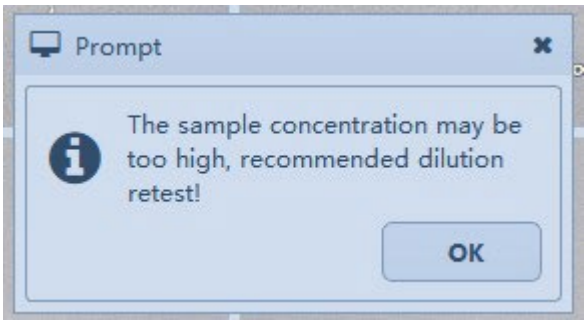


Fig. 4-22 Alarm for hematuria, pyuria

Alarm for Bacteriuria

If dry chemistry item LEU 3+, NIT+, or the number of bacteria is too high, the test status icon would turn orange, the system would alarm with "The sample may be bacteriurial" when cursor staying on the item in sample information list. The system would also alarm with "The sample may be bacteriurial" when user click 【Confirm review】.



Fig. 4-23 Alarm for bacteriuria

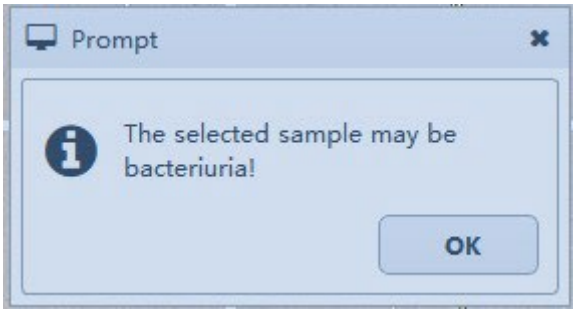


Fig. 4-24 Alarm for bacteriuria

4.3.6.4 Cancel review

For a sample having been reviewed, you can click <Cancel Review> to cancel review. You can do it in the <Sample Information> or Image Review interface.

⚠ NOTE

- If the sample has been reviewed, change is impossible.
- "Low Value" and "High Value" in the Result List respectively represent the minimum total number and maximum total number of a component in the images of all groups in the sample.

⚠ Warning

- The operator only can cancel the sample reviewed by himself, can not cancel the sample reviewed by others.
- The instrument can only identify a sample of a typical formed element,the subsidence is not ideal, microscope focal distance is not ideal, are likely to affect the recognition rate, so the doctor need to compare the microscope test result with the dry chemistry result, after that ,doctors can send the final report. Crystals and fungi can be easily misrecognized as cells, so be careful to check.
- In the process of recognition, if the formed element is at the edge of the image, the recognition of formed elements will be missing. When multiple formed elements are close to each other, our instrument will not recognize these formed elements and cause the false negative results,so doctors be required to

mark the formed elements manually.

- If there are to many mucus filaments and bacteria in the urine sample,these formed elements will affect the accuracy of the results,doctors must pay attention to this situation.
- For highly concentrated urine samples,we suggest the doctor to dilute the urine samples before testing.

4.3.6.5 Quick review

In the <Sample Information> interface, select the sample and click the right mouse button to show the context menu; <Confirm Review> will appear, and you can perform quick review and cancel review; this operation supports multi-choice. See Fig. 4-25.

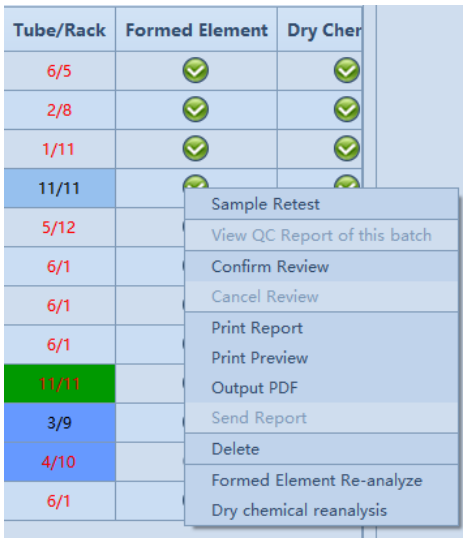


Fig. 4-25 Quick review interface

4.3.6.6 Print

- You can print the result of a reviewed sample. In the <Sample Information> interface, select the sample for printing, and click <Print Report>. Also, you can click <Print Preview> to preview the report.

- The instrument supports multiple report formats; you can select the report format in the <Setting> interface. See 4.8.5.

4.3.6.7 Send

- Send the test results to other terminal devices, such as LIS and HIS servers.
- In the <Sample Information> interface, select the record, and then click <Send Sample>.
- See 4.8.4 for Send settings.

4.4 Emergency sample

After the emergency sample is correctly put in the emergency position, you can create a new emergency sample at any time. Click the <Test> interface and click the <Emergency> button. After entering <Start No.> or <Bar code>, click <OK> to quickly create a new emergency sample; the system will give priority to the test of emergency sample. (See Fig. 4-26)

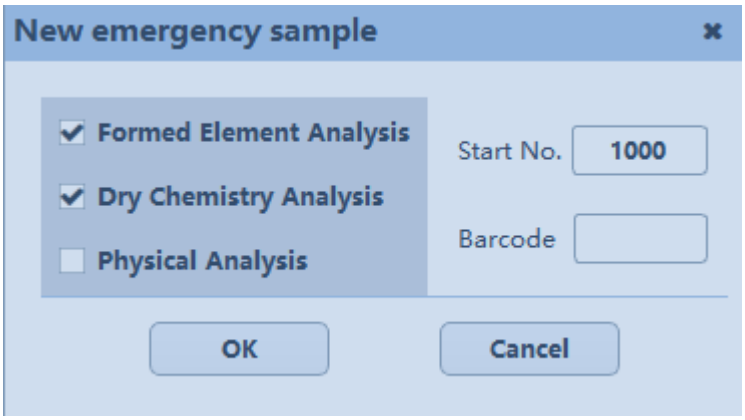


Fig. 4-26 New emergency sample

If a emergency test is required during normal test, operate according to the steps below:

1) Inject the emergency sample to a clean test tube; the sample volume should not be less than 3 mL.
2) Put the test tube in the emergency test tube rack.

3) Click the <Test> interface, and click the <Emergency> button.

4) Push the emergency test tube rack inward to the emergency position; the emergency position is provided with the magnetic self-locking function.

5) The instrument performs emergency test.

⚠ NOTE

To avoid damage of the instrument, do not take away the sample on the emergency test tube rack before emergency test is finished.

4.5 Retesting sample

1) Pull out the emergency rack and put in the sample for retest.

2) Push in the emergency position.

3) In the <Sample Information> interface, select the sample for retest; click the right mouse button, and click Sample Retest. See Fig. 4-27.

4) When sample retest is finished, take out the retested sample.

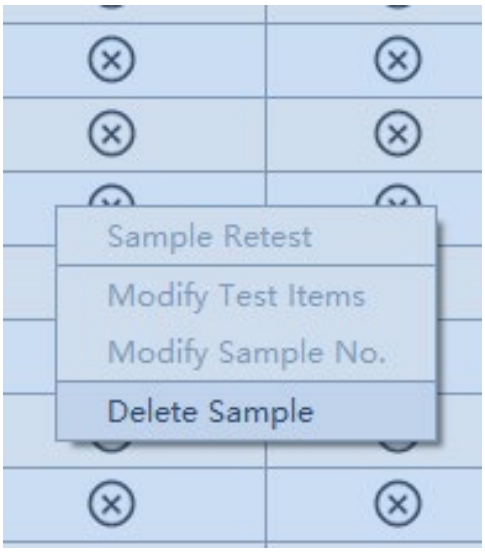


Fig. 4-27 Sample retest

4.6 Query history

◆ In the <Sample Information> interface, you can query the information of all samples by Test Date, Sample Type, Bar code, Patient Name, Admission No.,

etc. The number of samples that can be queried will be displayed. See Fig. 4-28.

- Click the sample to be viewed; the Sample Information bar and the Result bar will show the information and result of this sample. Also, you can enter, modify, review and print sample information.

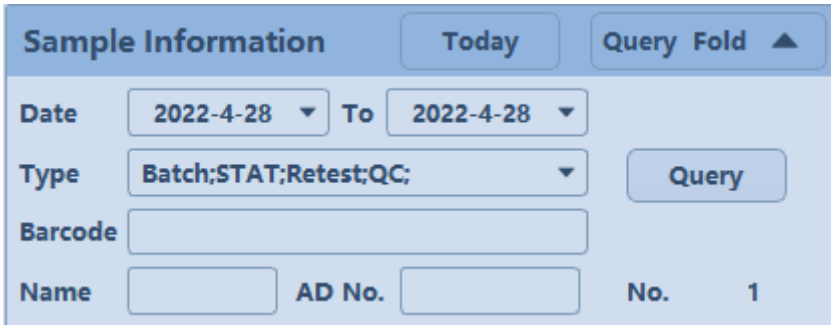


Fig. 4-28 Conditional query

4.7 QC

4.7.1 QC settings of formed elements

Navigate to <QC> and then click submenu----<QC> to enter the QC setting interface.See Fig. 4-29.

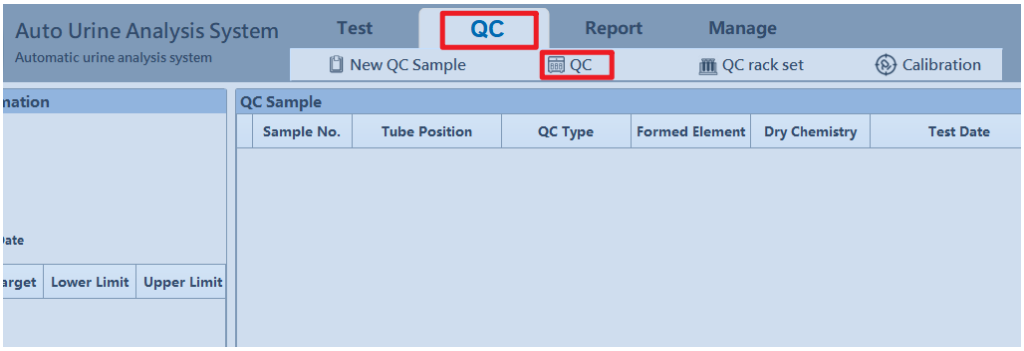


Fig. 4-29 QC settings of formed elements

Click <Formed Element> to enter Formed Element setting interface. Then fill <Batch No.>, <Type>, <Specification> and <Expiry Date> with the information from the package. At last click <add> to create a new item for Formed Element.See Fig. 4-30.

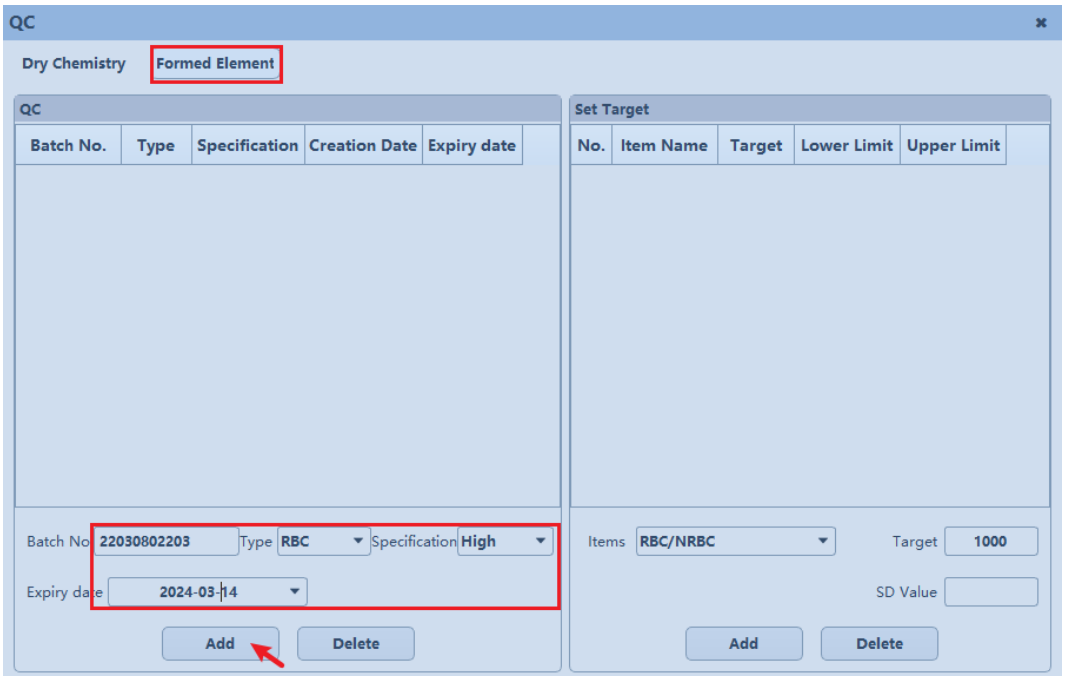


Fig. 4-30 QC settings of formed elements

Firstly, select the item of formed element. Then select corresponding items and fill the target value and SD value with the reference range of Urine Sediment Analyzer Quality Control in the package. At last click add to finish setting. See Fig. 4-31 .

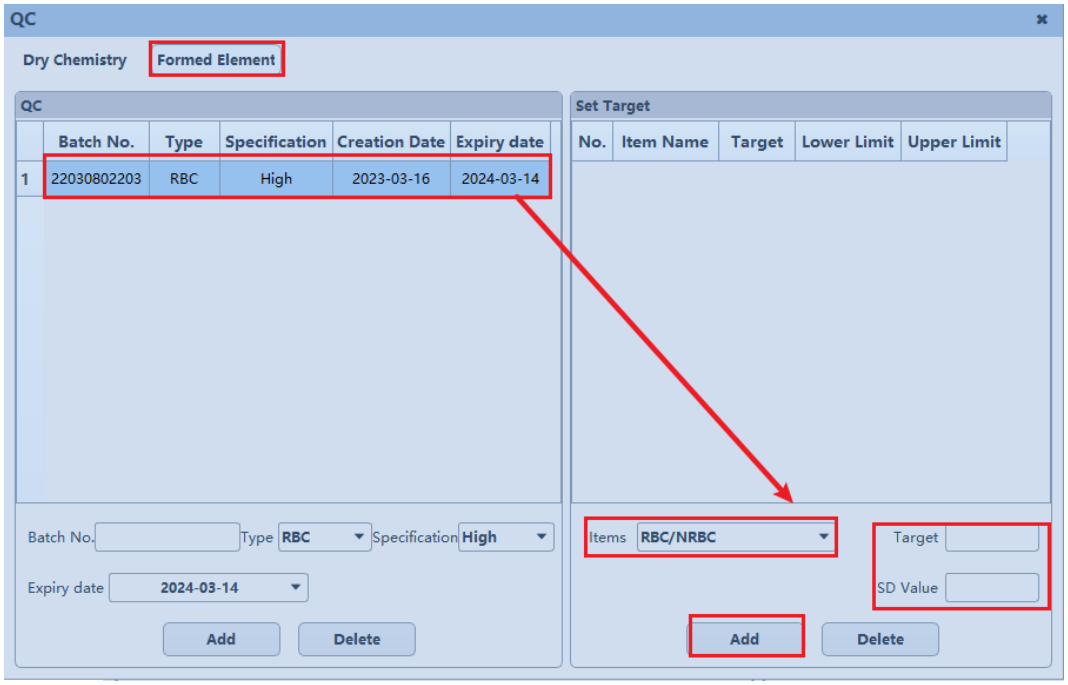


Fig. 4-31 QC settings of Formed Elements

4.7.2 QC settings of dry chemistry

Navigate to <QC> and then click submenu----<QC> to enter the QC setting interface. See Fig. 4-32.

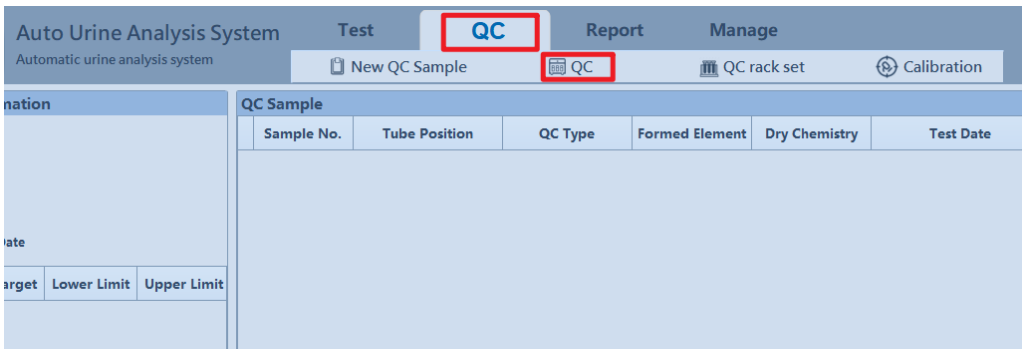


Fig. 4-32 QC settings of dry chemistry

Click <Dry Chemistry> to enter dry chemistry setting interface. Then fill <Batch No.>, <Name> and <Expiry Date> with the information from the package. At last click <add> to create a new item for Dry Chemistry QC procuct .See Fig. 4-33.

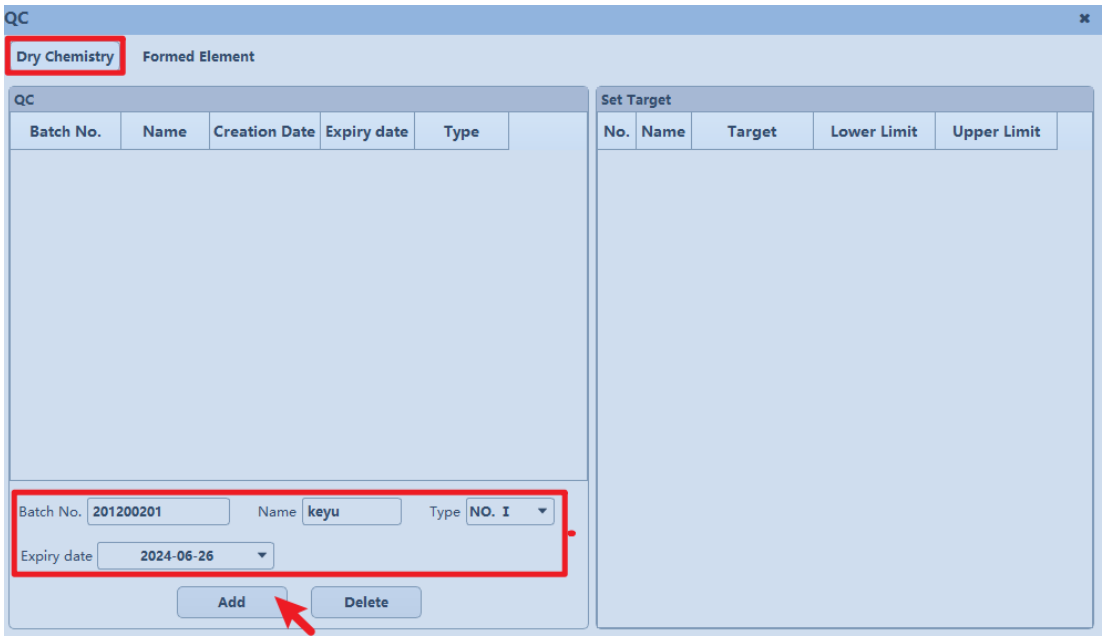


Fig. 4-33 QC settings of dry chemistry

Set target values for each test items according to the table of target values in the package of QC product.See Fig. 4-34.

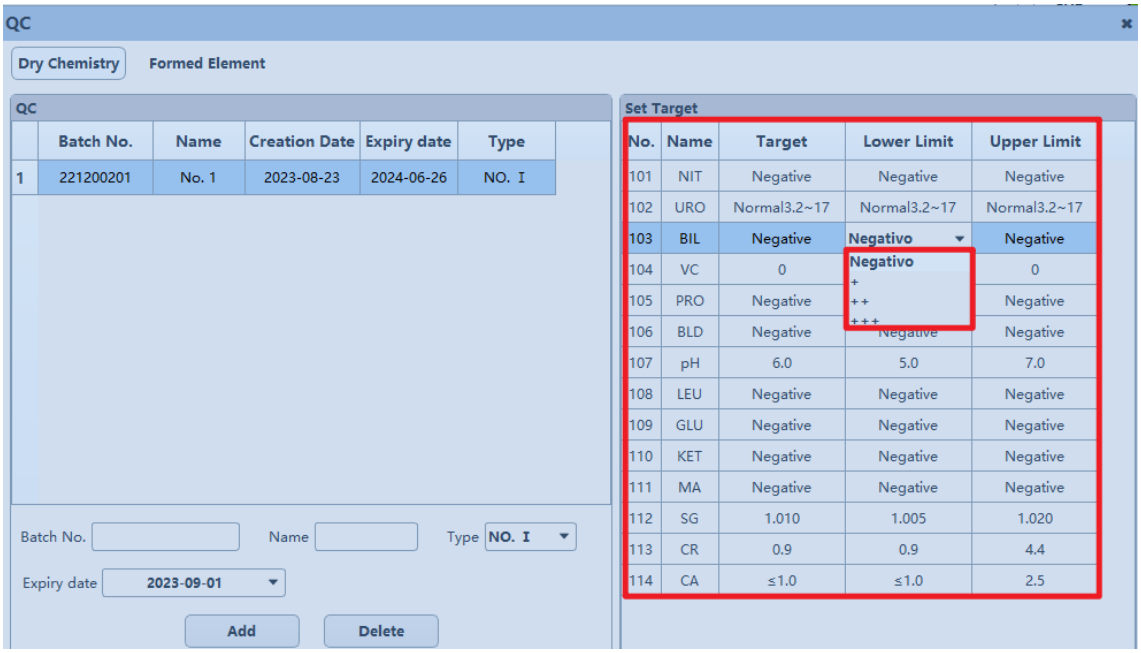


Fig. 4-34 QC settings of dry chemistry

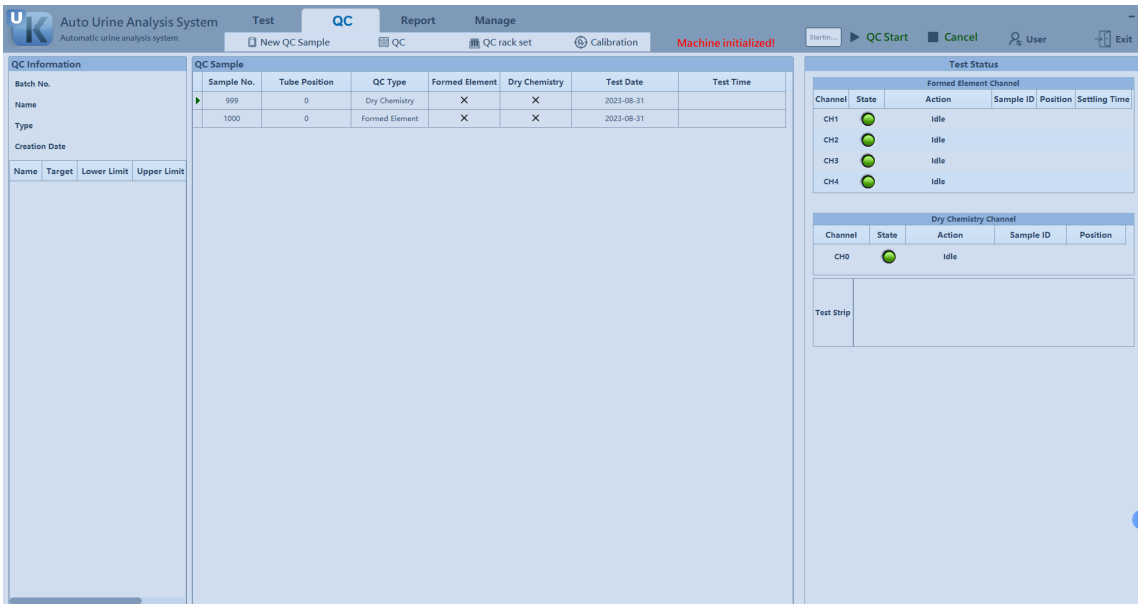


Fig. 4-35 QC interface

In the <QC> interface, as shown in Fig. 4-35. On the right side is the QC information, and on the left side is the current test status.

In the Sample Info List in the Review interface, select QC Sample; click the right mouse button to pop up the menu; select "View QC Report of this batch" to open the QC Report interface. (See Fig. 4-36)

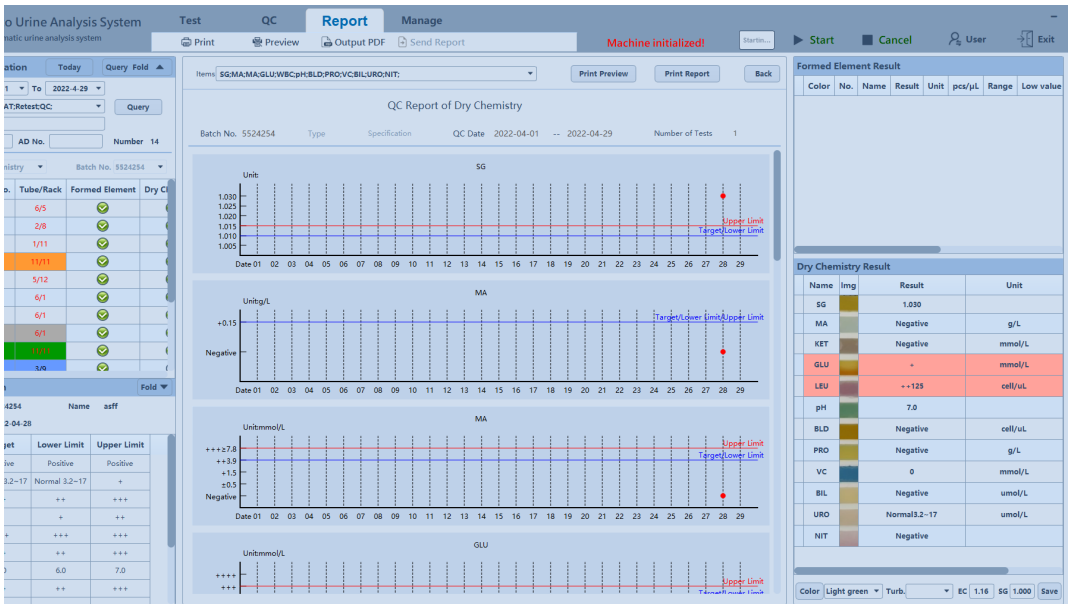


Fig. 4-36 QC report

"QC Information" at the lower left corner of the QC Report interface shows the QC information of the currently selected QC sample. "Items" in the middle can allow you to select the test item to show its QC chart information. QC samples can be queried by Batch No. within 31 days.

Quality control product sampling volume:

feces formed elements QC product--------2.5 mL per test

dry chemistry--------400 μL per test

4.7.3 The method to do formed elements quality control

Step 1: Take out the formed elements quality control product from the box, place the product on the sample holder and remove the cover. As shown in Fig. 4-37.

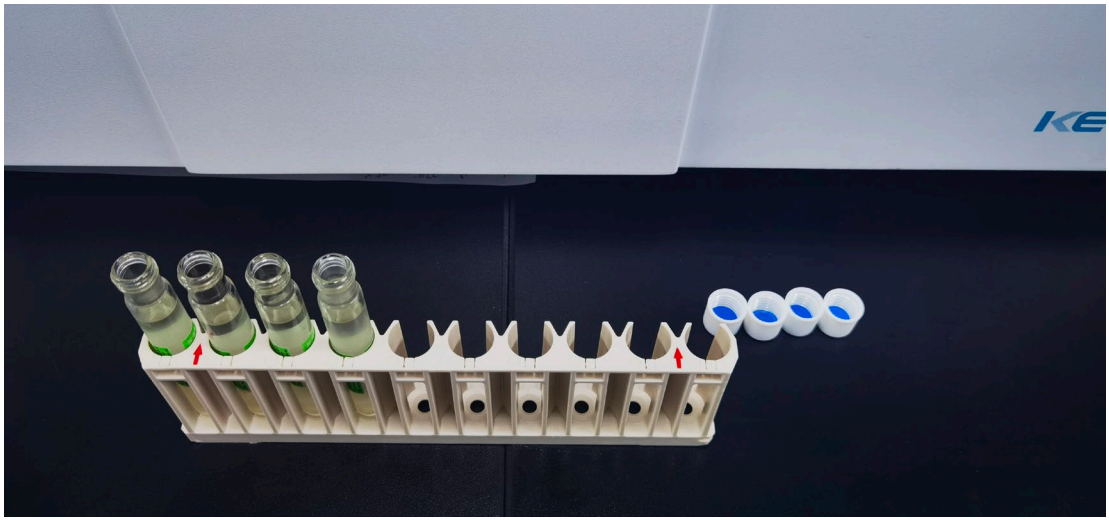


Fig. 4-37 Formed elements quality control product

Step 2: Place the sample holder on the tray. As shown in Fig. 4-38.



Fig. 4-38 Formed elements quality control product

Step 3: Create the new formed elements QC samples in the <QC> interface and click <QC Start>. As shown in Fig. 4-39/4-40.

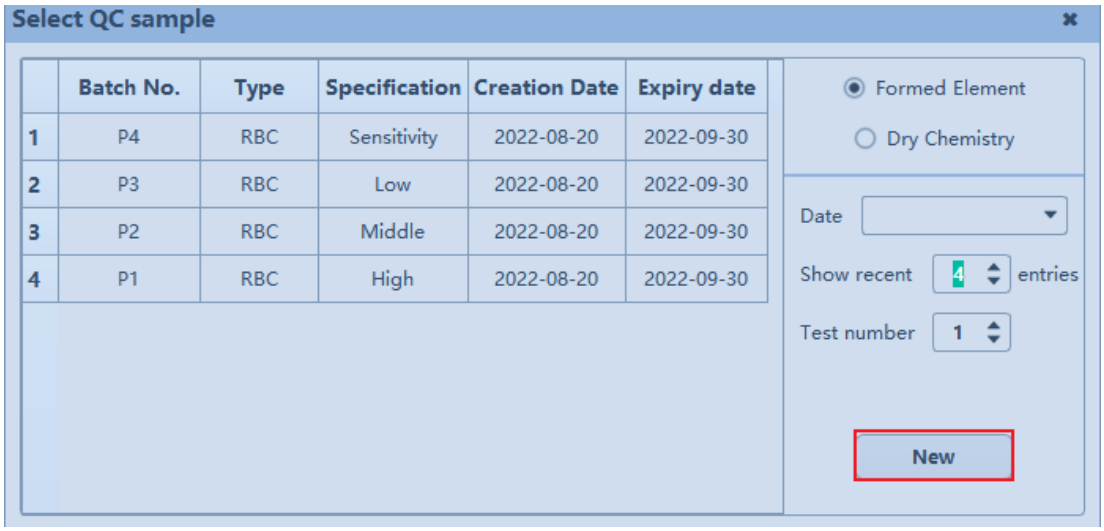


Fig. 4-39 Create new QC sample

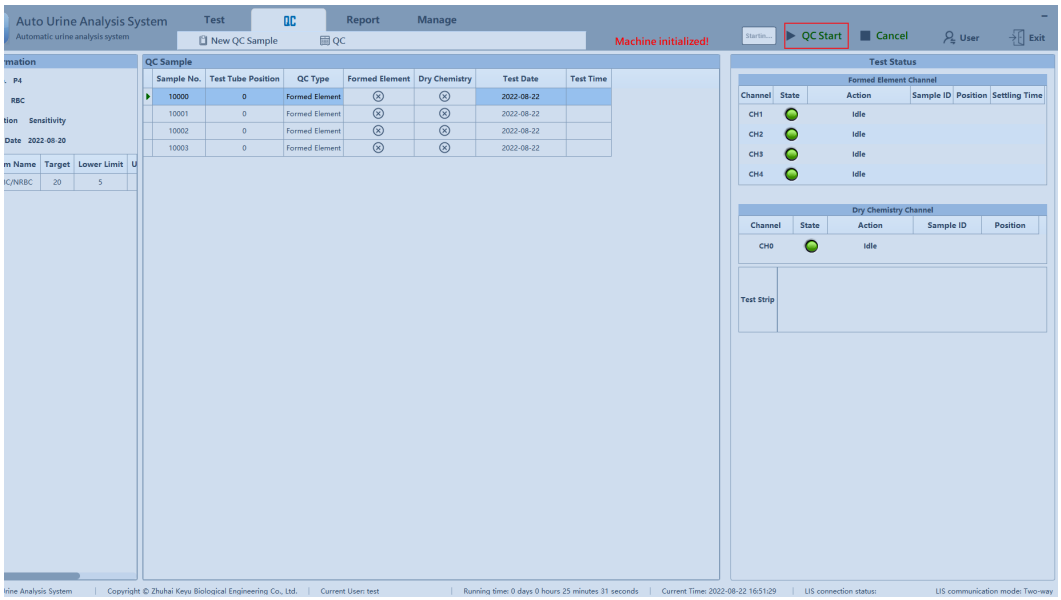


Fig. 4-40 Create new QC sample

Step 4: After the instrument finished the QC action, doctors can select the QC sample to check the QC results, right click the QC sample and select as shown in Fig. 4-41/4-42/4-43.

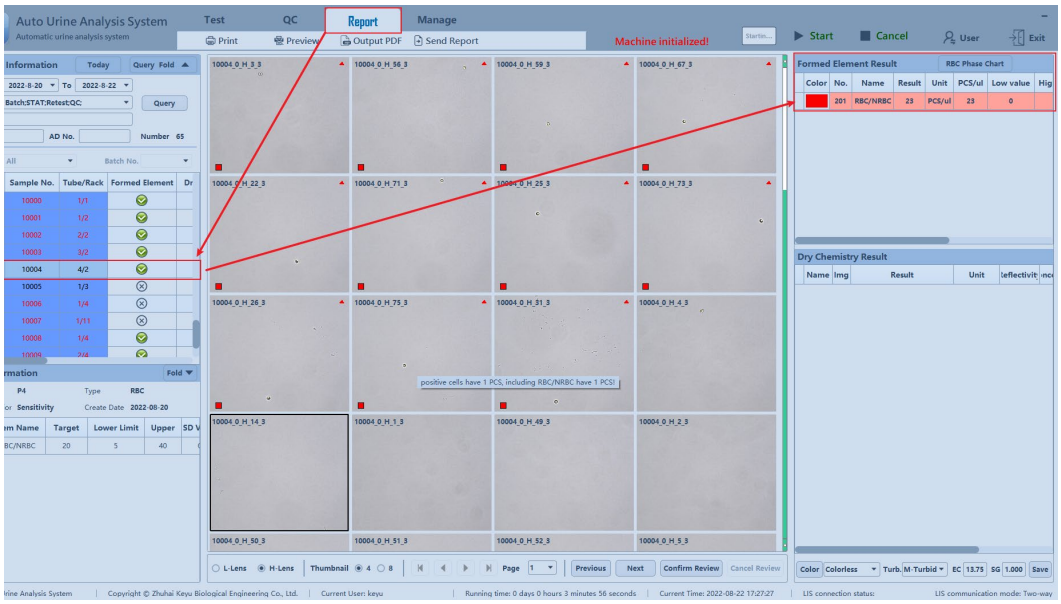


Fig. 4-41 Check QC result/report



Fig. 4-42 Check QC result/report

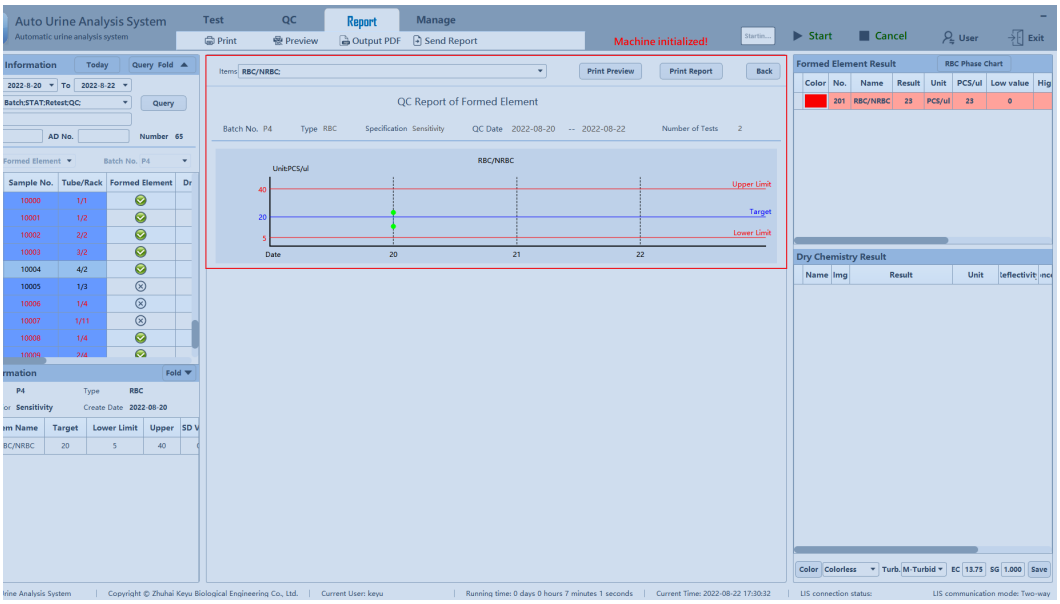


Fig. 4-43 Check QC result/report

4.7.4 The method to perform dry chemistry quality control

Step 1: Take out the dry chemistry quality control product from the box and remove the cap,As shown in Fig. 4-44.



Fig. 4-44 Remove QC product cover

Step 2: Use the pipette to extract about 1000 μL dry chemistry QC product to the test tube and place the tube on the sample holder. As shown in Fig. 4-45.



Fig. 4-45 Remove QC product cover

Step 3: Place the sample holder on the tray. As shown in Fig. 4-46.

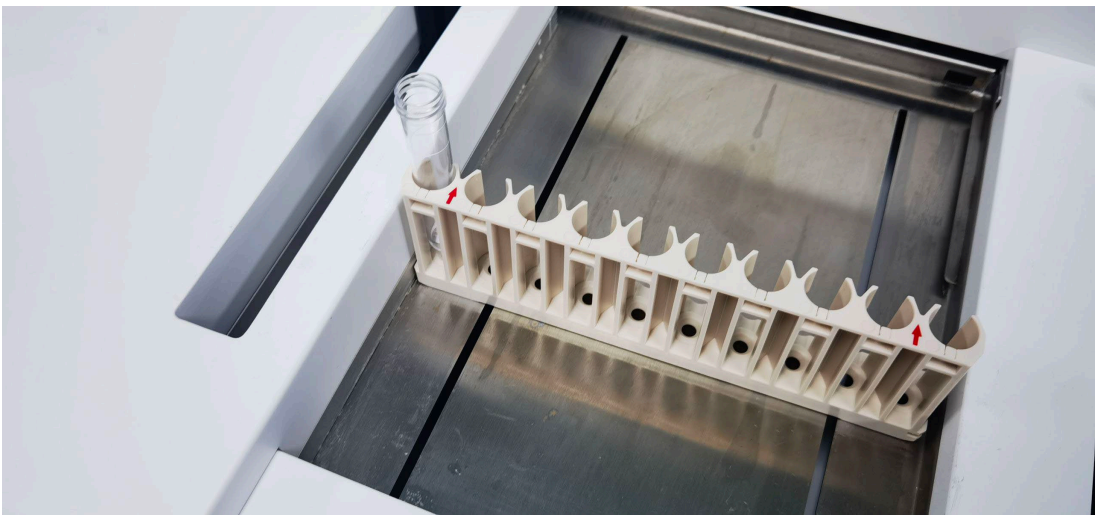


Fig. 4-46 Remove QC product cover

Step 4: Create the new dry chemistry QC samples in the <QC> interface and click <QC Start>. As shown in Fig. 4-47/4-48.

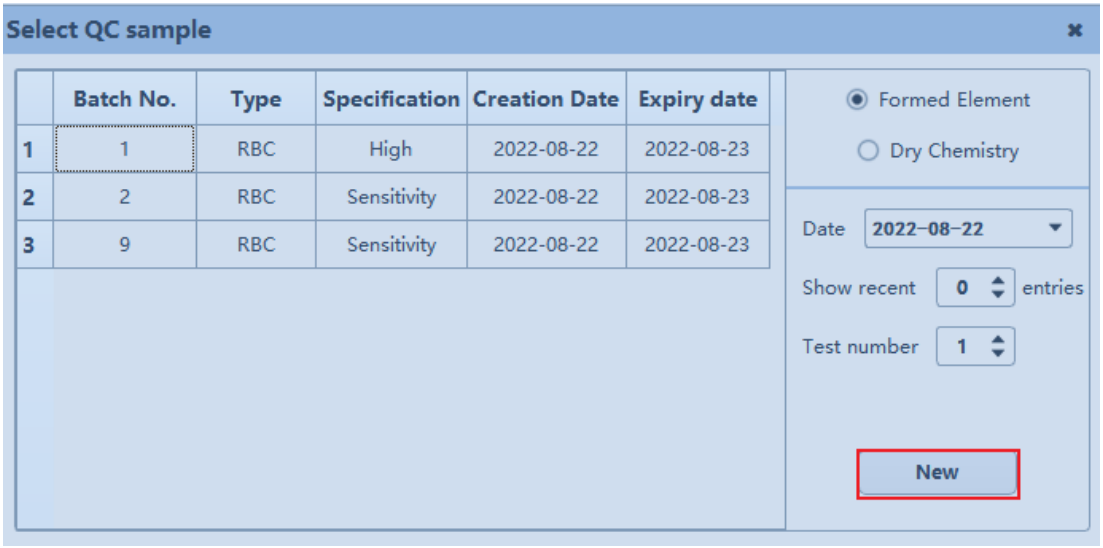


Fig. 4-47 Create new QC sample

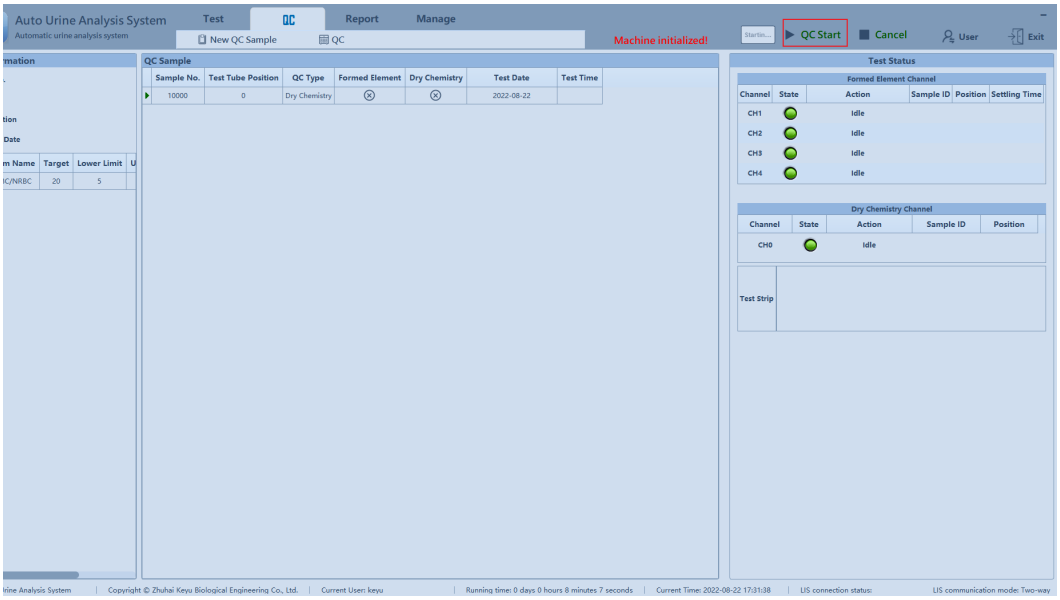


Fig. 4-48 Create new QC sample

Step 5: After the instrument finished the QC action, doctors can select the QC sample to check the QC results. As shown in Fig. 4-49.

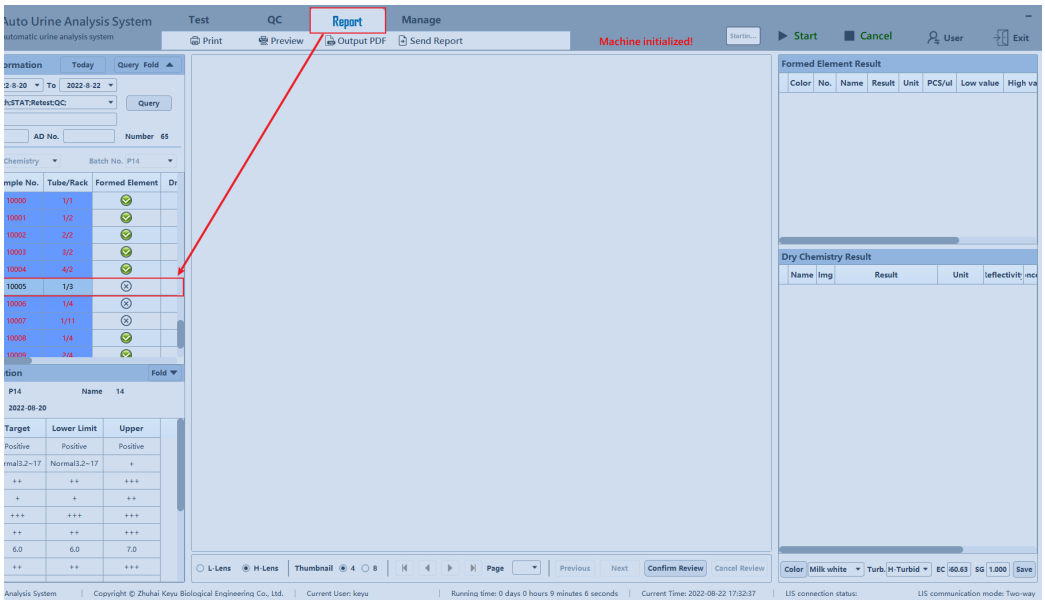


Fig. 4-49 Check QC result

Step 6: Right click the QC sample and select <View QC Report of This Report>. As shown in Fig. 4-50.

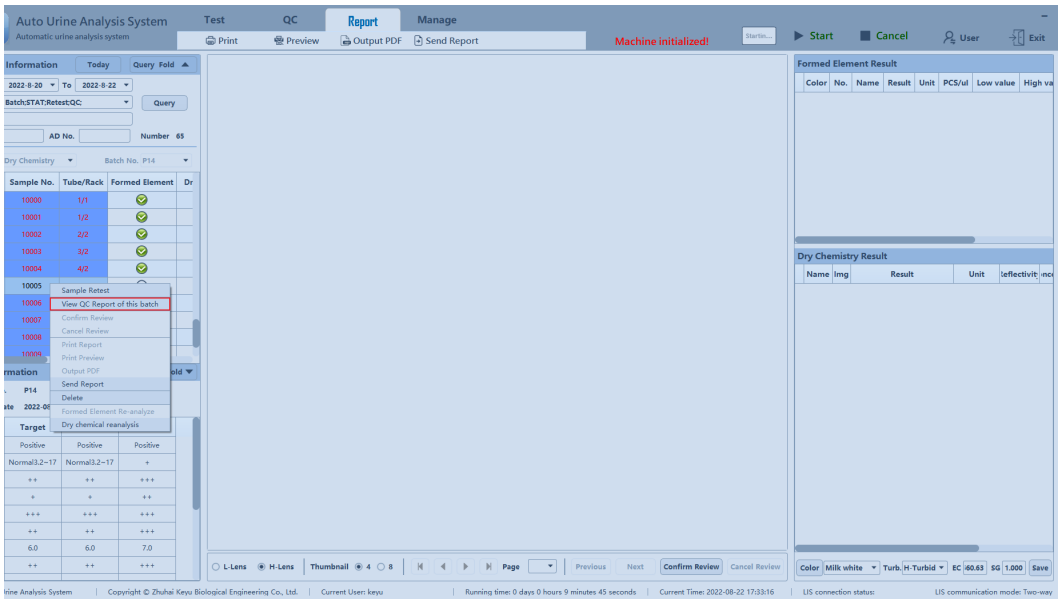


Fig. 4-50 Check QC report

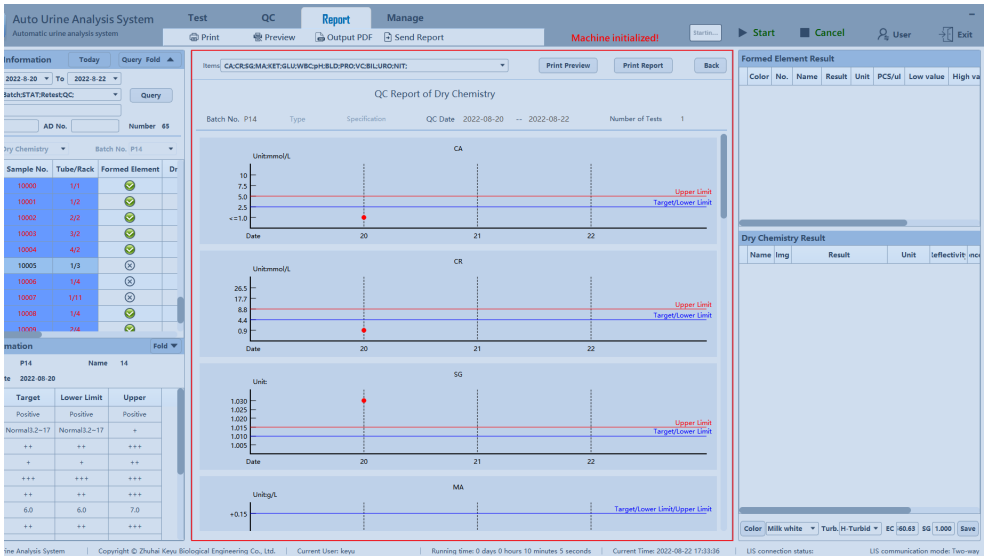


Fig. 4-51 Check QC report

4.7.5 Auto Creation of QC Sample for QC Tube Rack Setup

Click <QC> enter quality control interface, then click <QC Rack Set>, as the pic shown in Fig. 4-52. The user can tick on/off <Allow use of QCs beyond expiration date> and <Enable automatic creation of quality control samples>, then bind the types of QC with slots of the tube rack. Finally, close the setting interface and load the tube rack on the tray, click <QC Start> to carry out the QC test.

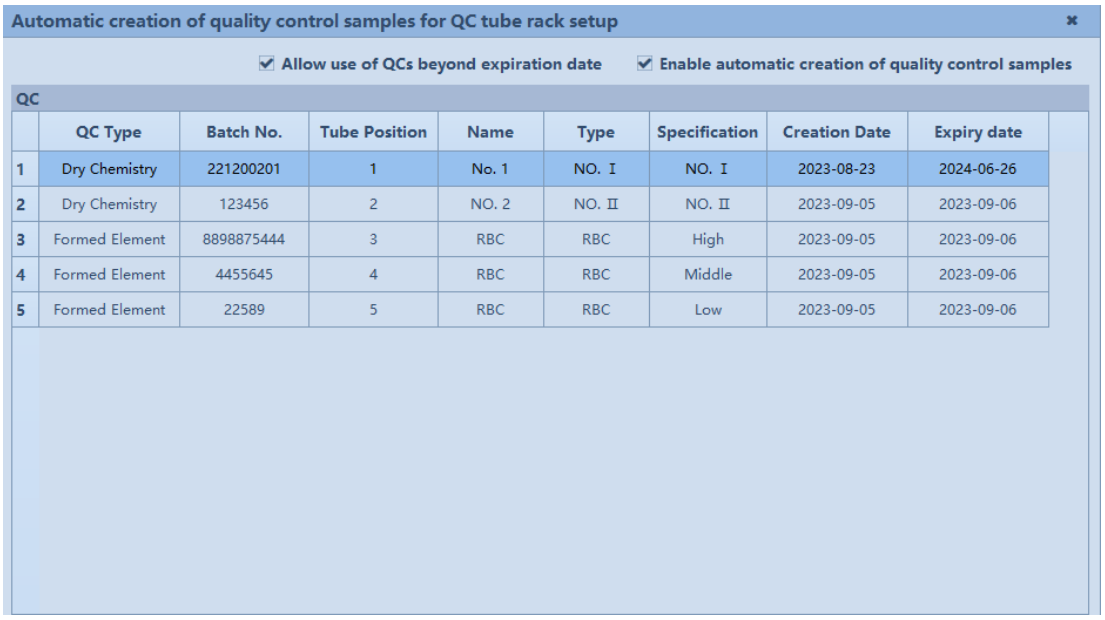


Fig. 4-52 Auto Creation of QC Sample for QC Tube Rack Setup

4.8 Formed element analysis calibration

4.8.1 Summary

In order to obtain accurate test result, we need to do the formed element analysis calibration, ensure the deviation correction factor under the specified conditions. To obtain accurate sample analysis results, the instrument should be calibrated when needed according to the steps.

Our instrument only support manual calibration method.

⚠ NOTE

- Only the administrator can do the calibration operation.
- The operator should use our company's calibrator and regent, and store and use the calibrators and reagents in strict accordance with the using instructions.

4.8.2 Calibration frequency

The instruments have been calibrated before they out of factory and do not require frequent calibration, instruments should only be calibrated in the following cases:

(1) Major components related to the analysis of formed elements have been replaced.
(2) When the instrument has not been used for a long time.
(3) The quality control data have a huge deviation.

⚠ NOTE

The test result can be used as a available data only after the calibration.

4.8.3 Calibration method

Calibration step:

(1) Place the calibration on the tube holder and test normally using Batch Start. Measure at least 5 times and record the results.

(2) Use calibration formula to calculate the new calibration coefficient. The calibration formula is as follows:

$$\text{New calibration factor} = \frac{\text{current calibration factor} * \text{reference value}}{\text{average value of measurement}}$$

For example: If the calibration reference value is 1114.4, the reference range is 687.5-1541.3, the current calibration factor is 110%.

Measure the calibration 6 times (n=6), and the results are 1035, 998, 1112, 1108, 1016, 983.

Index of repeatability CV=15%, meet the requirements, the average value of measurement is valid, if the measured value is outside the reference range, the result is invalid and cannot be used.

$$\text{New calibration factor}=\frac{\text{current calibration factor}*\text{reference value}}{\text{average value of measurement}}=117.6\%$$

If the calculated calibration factor outside the valid range (75%-125%), the calibration factor is invalid. In this situation, the operator need to look for the cause, then recalibrate and calculate the calibration factor.

(3) After obtaining the new calibration factor, enter the new calibration factor in the calibration factor box.

(4) After entering the new calibration coefficient, click the "Save" button. If the message "Save successfully" is displayed and the calibration date is updated to the current system date, the current calibration factor is valid. Otherwise, the current calibration factor is invalid.

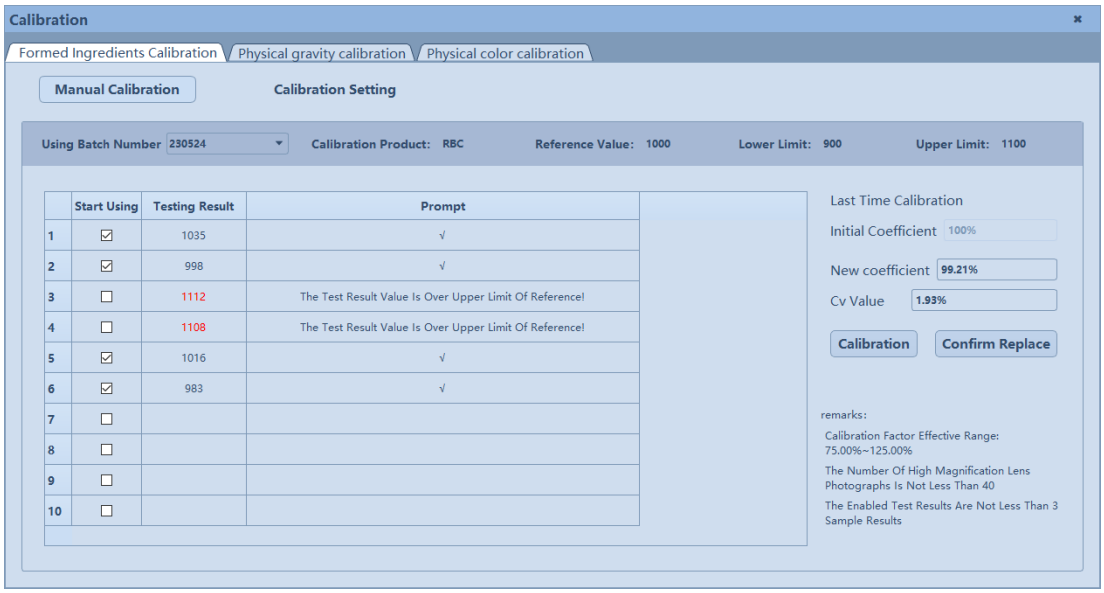


Fig. 4-53 Formed Ingredients Calibration

4.9 Physical color calibration

Calibrate the result of physical color by configuring the setting parameters. (mapping table method)

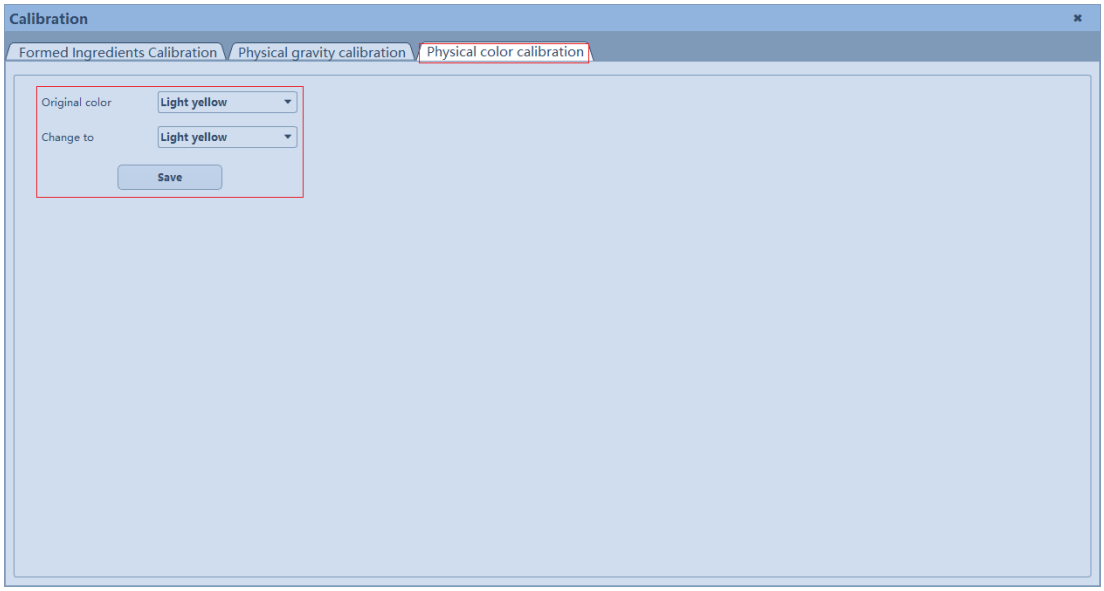


Fig. 4-54 Physical Color Calibration Diagram

4.10 Specific gravity calibration

Analyze the value of the specific gravity refractive index sent back from the lower computer and display it graphically. As pic shown in Fig.4-55.

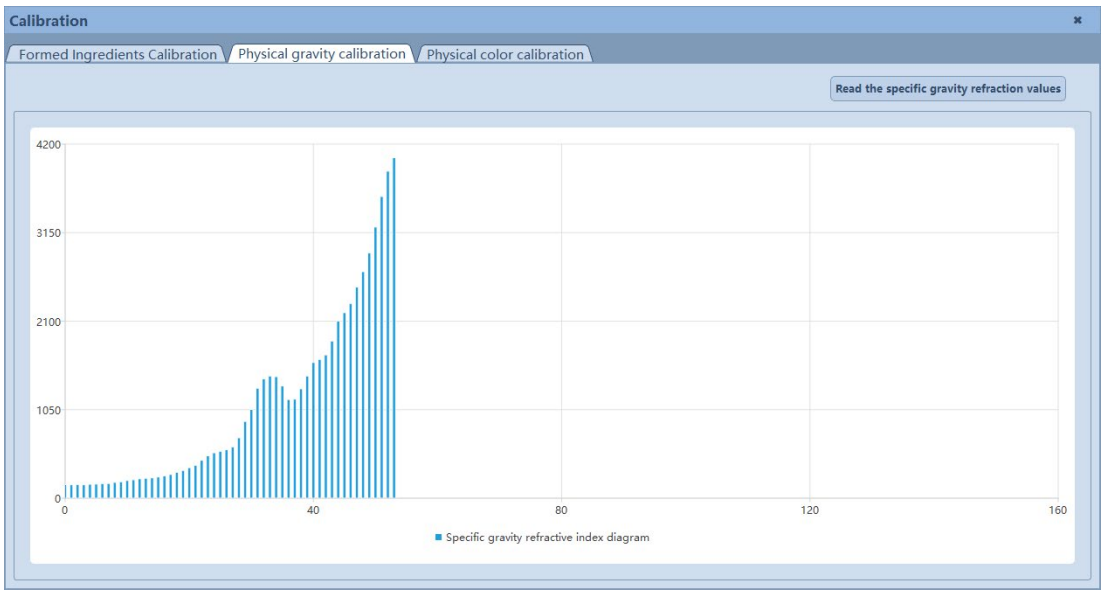


Fig.4-51 SG Refractive Index Diagram

4.11 Settings

4.11.1 Set shoot parameters of microscopy: You can set Number of Views, Shooting Layers and Enable Channels. Set Strip Type: set KU-12B or KU-14A as the type of test strip currently used. (See Fig. 4-56)

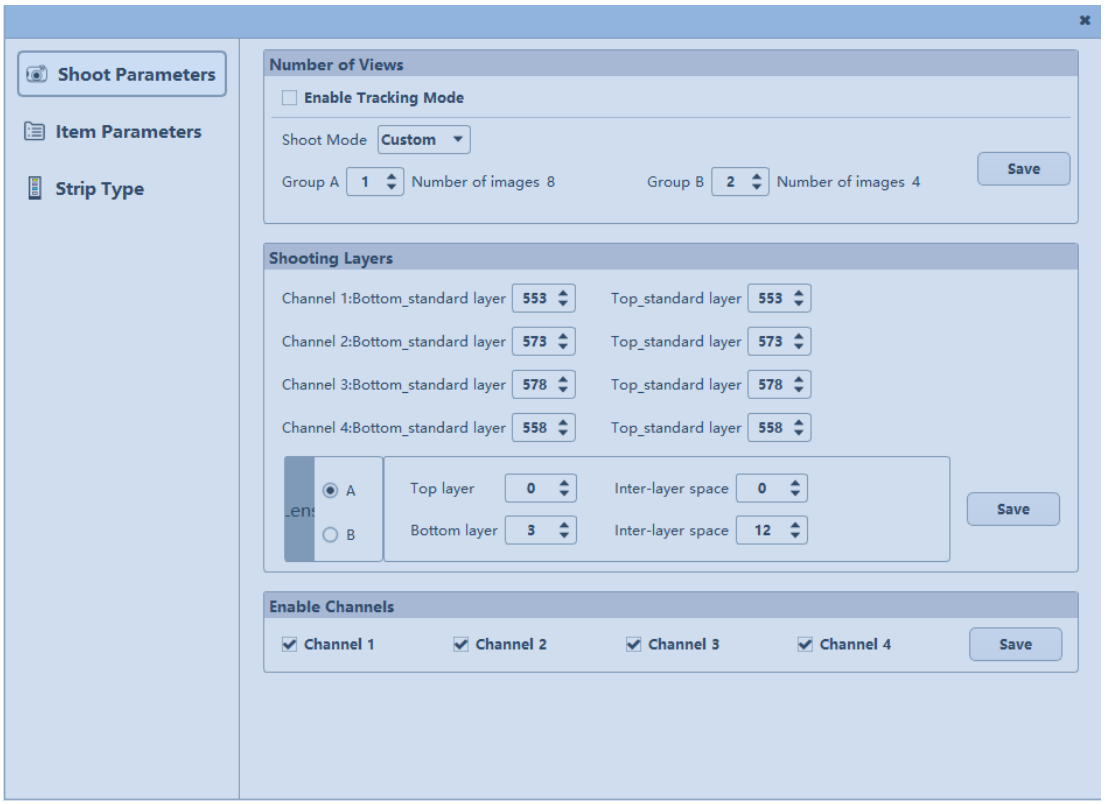


Fig. 4-56 Shoot Parameters Interface

4.11.2 Set item parameters of formed elements: Select items through the drop-down list, click <Add> or <Delete> to add or delete items, and after setting remember to click <Save>. (See Fig. 4-57)

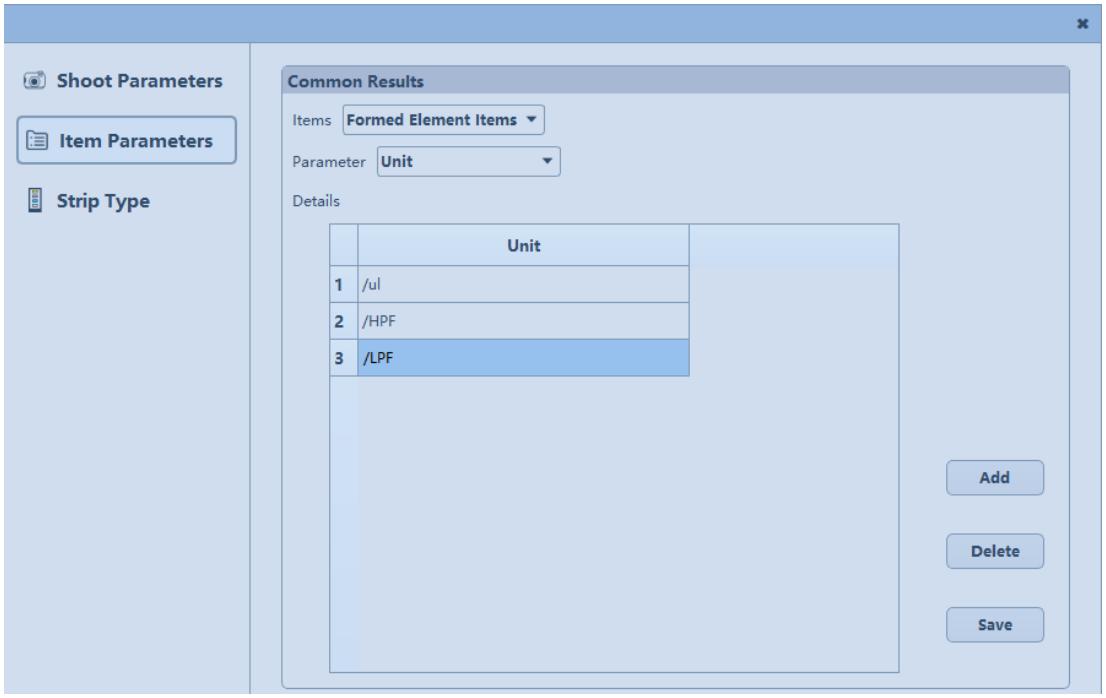


Fig. 4-57 Items Parameters Interface

4.11.3 Set test strip: The current test strip type can be selected by the drop-down list, click <Set as current test strip> to set the type. (See Fig. 4-58)

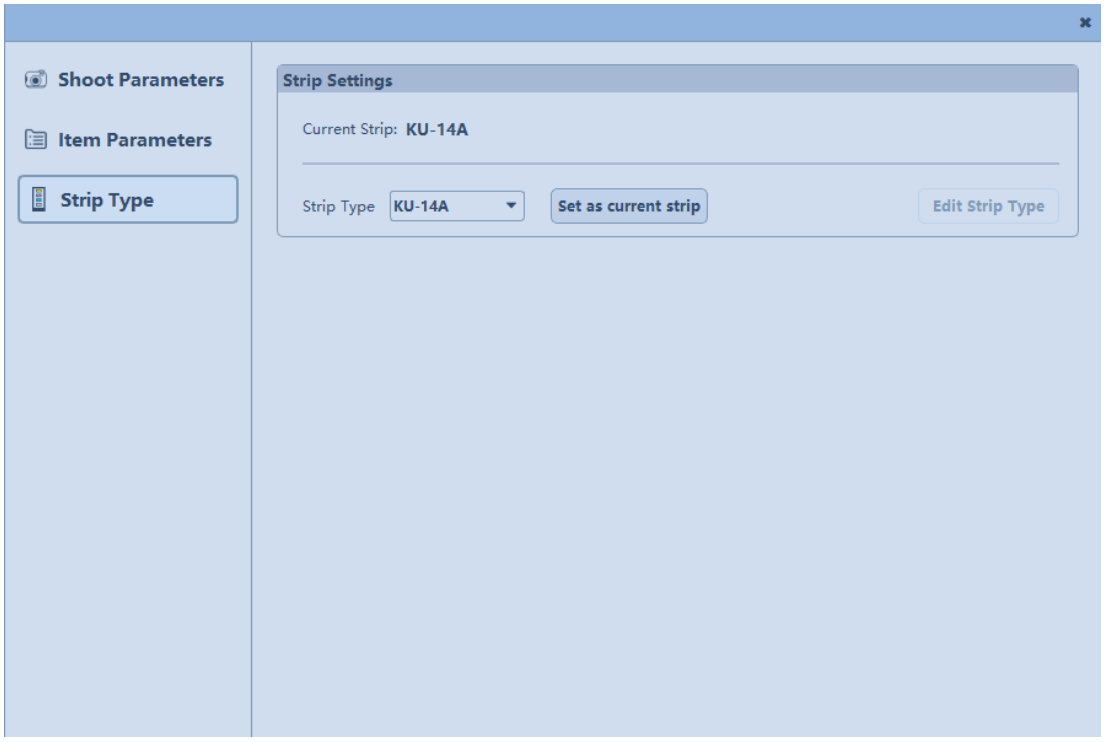


Fig. 4-58 Strip Type Interface

4.12 System management

4.12.1 Set User Manage: For management of operators, submission departments and submission physicians. See Fig. 4-59.

1) User Manage

Username —— The name you need to use for login.

Name —— The operator's name.

Permission —— There are two user groups for system settings: "Advanced User" and "General User". Advanced user can use all functions, and general user cannot perform system settings and does not have the permission of adding/deleting users.

- Advanced user can add, modify and delete users.
- General user can modify his/her own password.

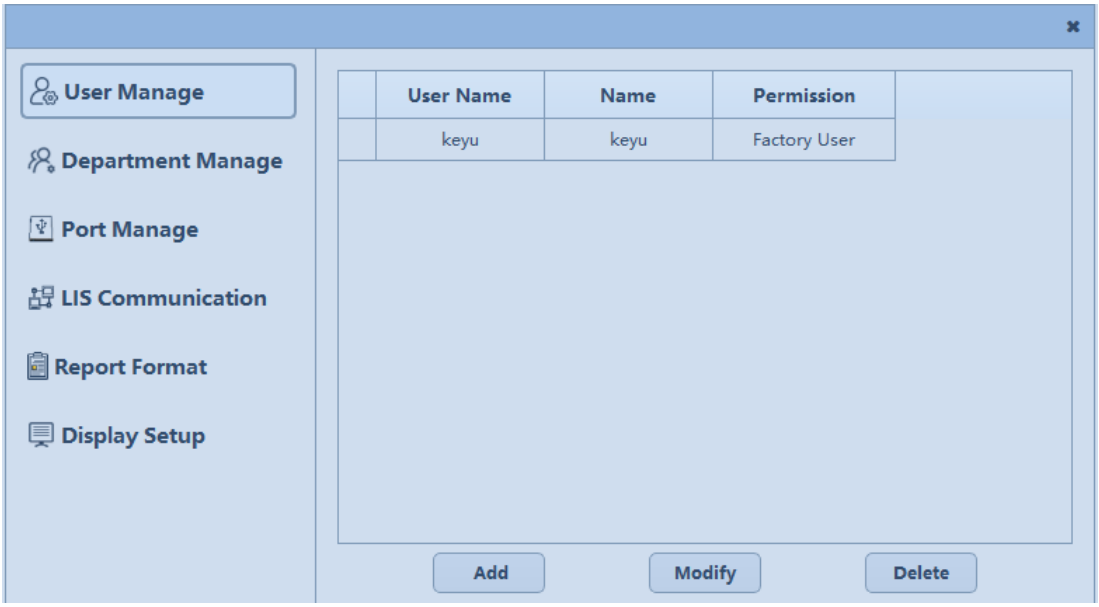


Fig. 4-59 User Manage Interface

4.12.2 Department Manage: For adding of departments and doctors. See Fig. 4-60.

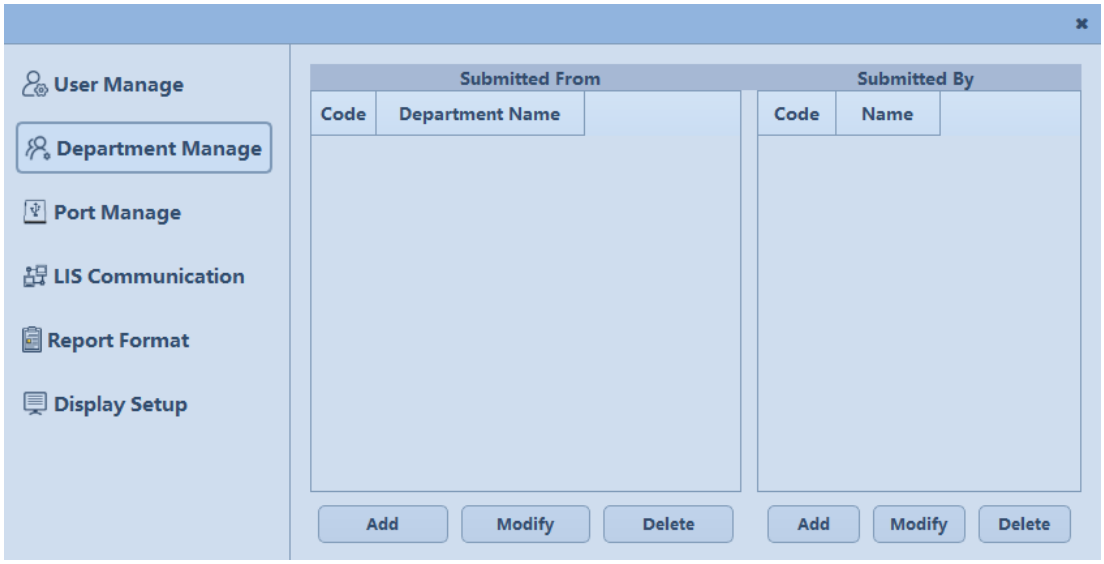


Fig. 4-60 Department Manage Interface

4.12.3 Port Manage: For setting of communication ports of upper and lower computers. See Fig. 4-61.

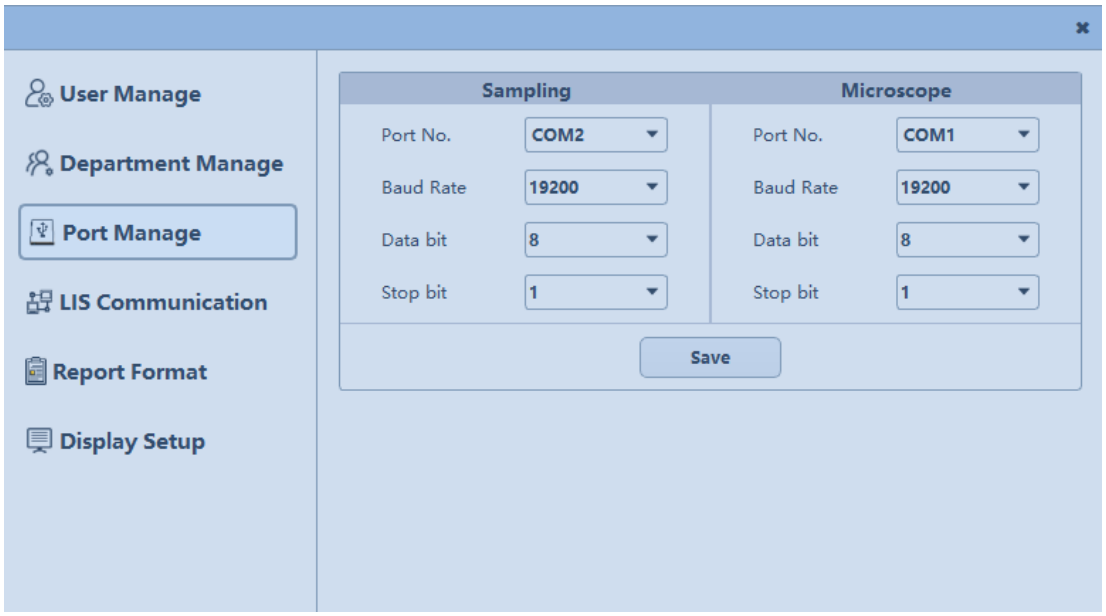


Fig. 4-61 Port Manage Interface

4.12.4 LIS Communication Manage: For setting information related to LIS connection. See Fig. 4-62.

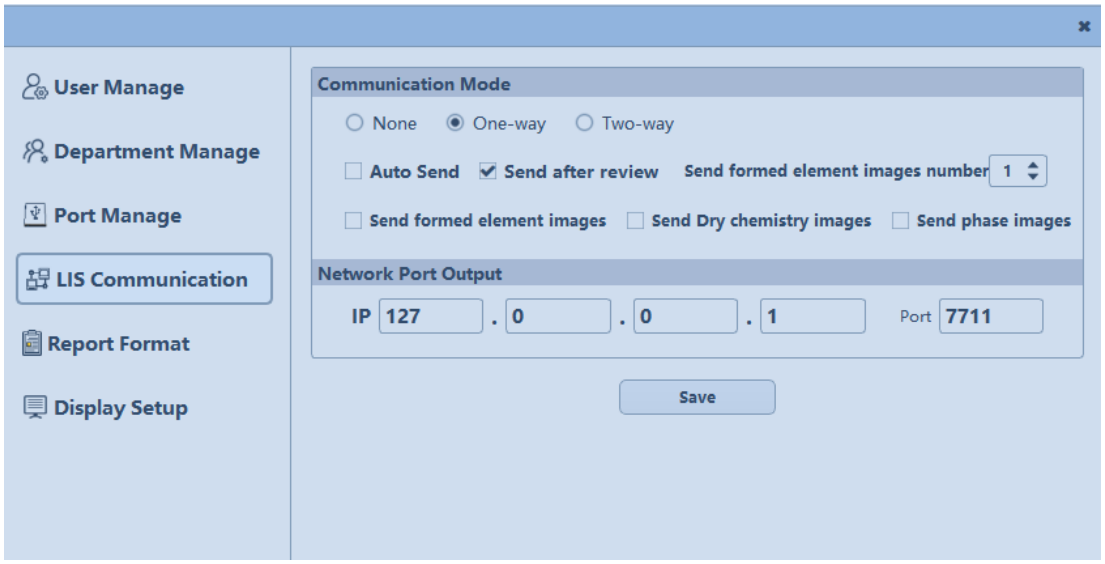


Fig. 4-62 LIS Communication Interface

4.12.5 Report Format: For setting the format of printed report. See Fig. 4-63.

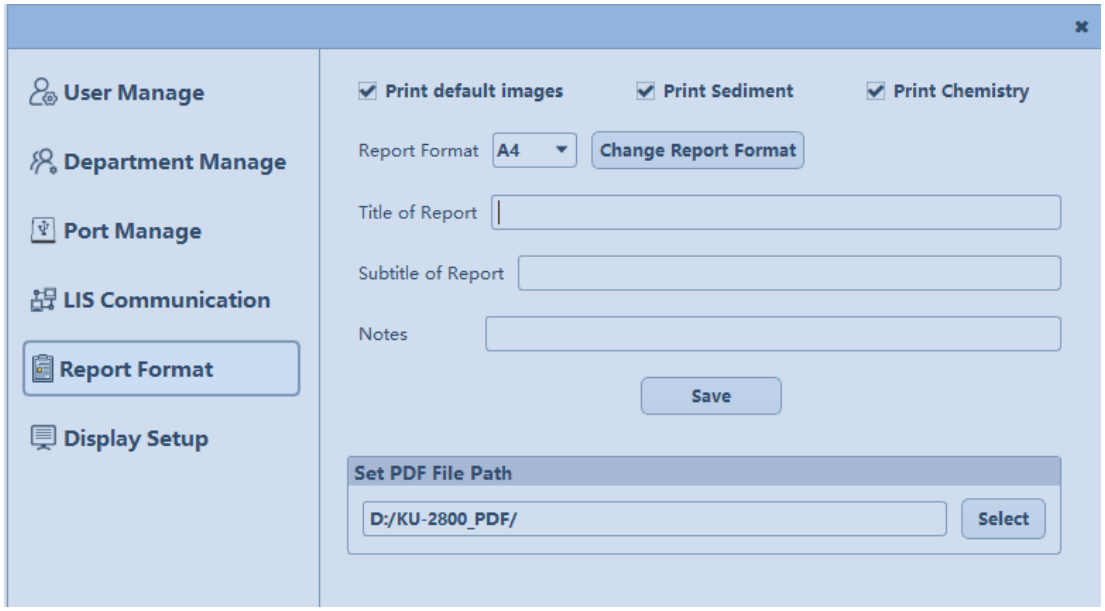


Fig. 4-63 Report Format Interface

4.12.6 Set Language

Select the language mode from the drop down menu,then click <OK> to save the setting.

As the pic shown in Fig.4-64:

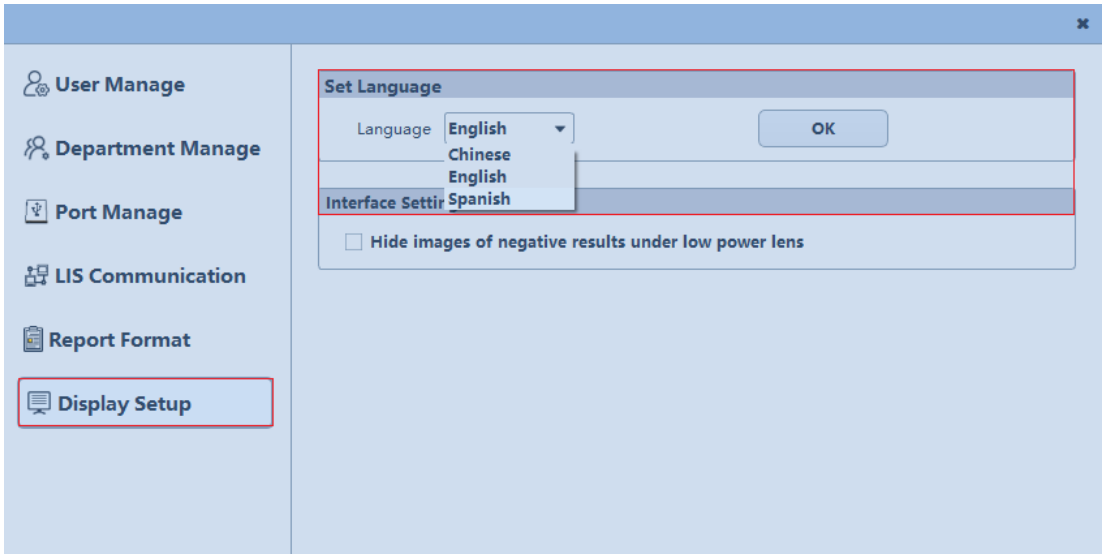


图 4-64 Language Setting Interface

Chapter 5 Maintenance of instrument

The product is a precision electronic optical instrument. To ensure normal operation of the instrument, periodic maintenance is very important. If the instrument malfunctions, it shall be repaired by professional maintenance personnel with AGAPPE's authorization or be sent to the service department designated by AGAPPE for repair.

Precautions:

1. Prior to operation, carefully read the User Manual for this instrument and the instructions for use of urine reagent strips;
2. The instrument must be kept clean to maintain good operation;
3. If the instrument is left unused for a long period of time, a soft cloth or plastic sheet should be used to cover it. Before next use, first check with QC liquid;
4. Do not disassemble the instrument.

⚠ WARNING 

- Please wear protective gloves to prevent bacterial infection.
- When disposing of liquid waste, used consumables, gauze, cotton swabs and gloves, the local regulations must be observed.

5.1 Maintenance items

Daily maintenance

Item	Interval
Clean the test strip compartment	Daily
Clean the waste strip box	Daily
Dispose of liquid waste	Daily

Periodic maintenance

Item	Interval
Clean the test strip carousel	Weekly
Clean the counting cell	Weekly
Channel maintenance	Twice a week

5.2 Daily maintenance

5.2.1 Clean the test strip compartment

Articles required for cleaning: Big-size air pump, tissue.

Step 1: Move away the test strips

1. Loosen the locking screws and open the cover of test strip compartment (see Fig. 5-1).
2. Take the test strips out of the test strip compartment (see Fig. 5-2).
3. Wrap the test strips with a cover to protect them from dirt.

⚠ NOTE

Do not touch the pad on the test strip; otherwise, skin surface lipid may adhere to the test strip, affecting the test result.

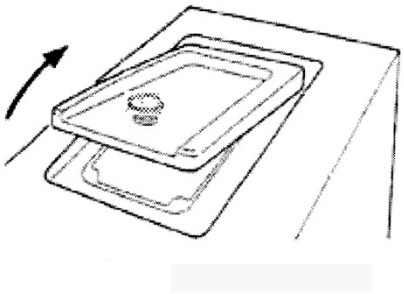


Fig. 5-1

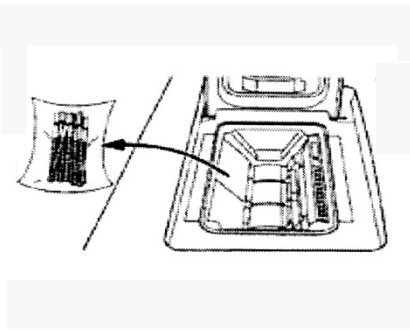


Fig. 5-2

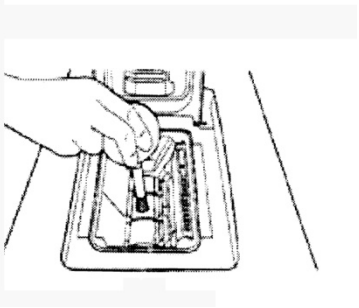


Fig. 5-3

Step 2: Clean the test strip compartment

1. Use the big-size air pump to clean the test strip compartment (see Fig. 5-3).
2. Put the test strips back to the test strip compartment.
3. Close the cover of test strip compartment.

Step 3: Clean the test strip delivery component

1. Open the side cover of the instrument.
2. Open the cover of test strip compartment rightward (see Fig. 5-4).
3. Open the cover of test strip delivery component.
4. Use the big-size air pump to clean the track area (see Fig. 5-5).
5. Close the cover of test strip delivery component.

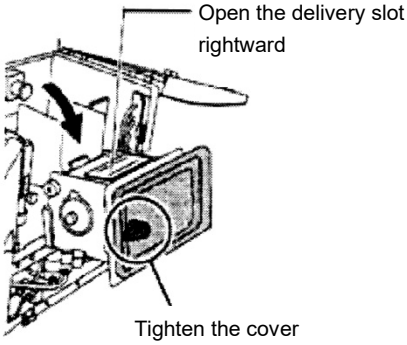


Fig. 5-4

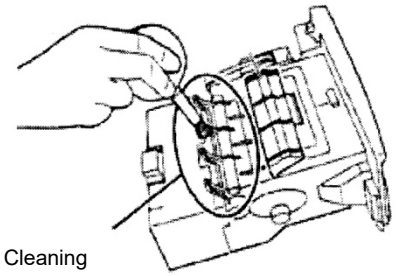


Fig. 5-5

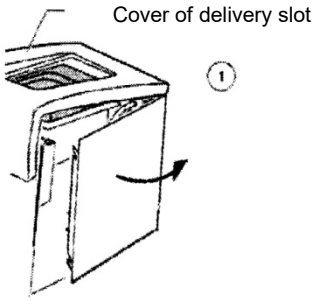


Fig. 5-6

Step 4: Finish cleaning

1. Reinstall the test strip compartment.
2. Close the cover of test strip compartment.
3. Open the door panel in the direction shown in Fig. 5-6.

5.2.2 Clean the waste strip box

The waste strip box can accommodate 500 test strips. Remove the used test strips in the box and disinfect the waste strip box.

⚠ WARNING

Please wear protective gloves to prevent bacterial infection.

⚠ NOTE

When disposing of liquid waste and gloves, the local regulations must be observed.

Articles required for cleaning: Alcohol, cotton cloth and protective gloves.

Step 1: Disposing of used test strips

1. Confirm that the instrument is in standby.
2. Open the waste strip box (see Fig. 5-7), and remove the test strips inside it.

Step 2: Disinfect the waste strip box

1. Use alcohol to clean the waste strip box; then wash with water (see Fig. 5-8).
2. Use the cloth to wipe dry the waste strip box.

Step 3: Reinstall the waste strip box

Install the waste strip box back to the original position (see Fig. 5-9)

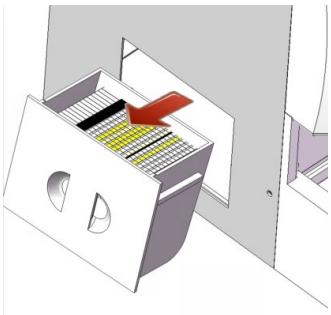


Fig. 5-7



Fig. 5-8

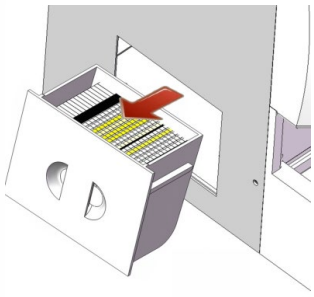


Fig. 5-9

5.2.3 Disposing of liquid waste

Dump the liquid waste after finishing every day's tests.

⚠ WARNING 

Please wear protective gloves to prevent bacterial infection.

⚠ NOTE

When disposing of liquid waste and gloves, the local regulations must be observed.

Articles required for cleaning: Alcohol, protective gloves.

Step 1: Disposing of liquid waste

1. Confirm that the instrument is in standby.
2. Remove the cap on the liquid waste bottle.
3. Dump the liquid waste.

Step 2: Clean the liquid waste bottle

Use alcohol to clean the liquid waste bottle and wash it with water.

Step 3: Assemble the liquid waste bottle

Install the cap back to the liquid waste bottle and tighten it.

5.3 Periodic maintenance

5.3.1 Clean the test strip carousel

During continuous test, residual sample liquid may adhere to the test strip carousel, obstructing smooth running of the carousel. Therefore, the test strip carousel should be cleaned on a weekly basis.

Articles required for cleaning: Alcohol, cloth, protective gloves.

⚠ WARNING 

Please wear protective gloves to prevent bacterial infection.

⚠ NOTE

When disposing of liquid waste and gloves, the local regulations must be observed.

Do not use any organic solvent for cleaning since it will cause deformation or discoloration of the test strip carousel.

Step 1: Cut off the power supply

After confirming that the instrument is in standby, turn off the system and cut off the power supply.

Step 2: Open the housing for strip selector of the instrument

1. Open the left-side housing of the instrument.
2. Turn the strip selector by 90°.

Step 3: Clean the test strip carousel

1. Use a clean cloth to remove dust and other dirt on the surface of test strip carousel.
2. After cleaning, restore all parts to their original positions.

⚠ NOTE

Do not use hot water; otherwise, the test strip carousel could get damaged. Use a cloth to wipe it dry.

Do not scratch the test strip carousel; otherwise, the operation of test strips could be affected.

5.3.2 Clean the counting cell

When the surface of counting cell is dirty, it is need to wipe it clean. Click <Wipe Position>; the objective will turn to the wiping position. Then use cotton swabs to wipe off the stains on the surface of counting cell. Close the maintain window; the objective will reset automatically.

⚠ NOTE

Focusing is suggested at least once a week. When images become unclear, focusing should be carried out immediately.

5.3.3 Channel maintenance

Perform maintenance at least twice a week

Step 1: Manually pour the maintenance solution into the sample collection cup (approximately 10 ml),Insert the tube filled with concentrated cleaning solution into the emergency position

Step 2: Click <Maintain> → <Reagent> → <Channel Maintenance> ,then the reagent wil be sucked into the channel 1,2,3 and 4 automatically

Step 3: After about 5 minutes,the instrument automatically flush maintenance solution with diluent solution, prompting "Maintenance Finished”.

5.4 Maintenance when the instrument is left unused for a long period of time

Articles required for cleaning: cleaning solution, diluent, sample test tube, cotton swab, purified water, alcohol, cleaning brush, tissue.

Step 1: Carry out daily maintenance

Clean the test strip delivery slot, and dispose of the used test strips out of the waste strip box.

Step 2: Discharge cleaning solution and diluent

1. Confirm that the instrument is in standby.
2. Pull out the pipeline pipe on the cleaning solution bottle.
3. Exit the software; turn off the system and restart it.
4. Cleaning solution will be discharged from the drain pipe.

Step 3: Dump the cleaning solution and diluent

Empty the cleaning solution bottle and diluent bottle.

Step 4: Dump the liquid waste

1. Dump the liquid waste out of the liquid waste bottle.
2. Clean the drain pipe and put it back.

Step 5: Put away the power cord

Disconnect the power cord of the instrument from the socket.

5.5 Check and replacement of pipe

When priming the instrument, check whether the pipe leaks. In case of liquid leakage, please contact our After-sales Service Department or agency for repair or replacement of corresponding parts.

5.6 Replacement of fuse

1. Carry out preparatory work before replacement. Cut off the power supply, and wear insulating gloves.
2. Start replacing the fuse. Remove the damaged fuse, and replace it with a new fuse. Model of fuse used on the instrument: F2AL250V.
3. Test the fuse. After replacing the fuse, use a test pencil to test the wiring voltage condition at both ends of the fuse holder. If voltage exists at both ends, it indicates the replacement is OK.
4. Power on the instrument.

5.7 Replacement of diluent and cleaning solution

(1) Take out the diluent pipe and cleaning solution pipe from the old bottle and insert the pipes into the new bottles, make sure the pipe head to the bottom of the bottle.

(2) Remove the old reagent card and insert the new reagent card, click <Read reagent volume> to read the volume and the time to make sure the reagent card is valid.

(3) Click <Perfusion> to make sure the diluent fill all the liquid pipe.

Chapter 6 Troubleshooting

This chapter gives the list of common faults of the instrument and corresponding solutions. When the instrument malfunctions, the operator can perform troubleshooting according to the alarm message generated by the instrument.

If you cannot solve the problem or needs more technical support, please call our After-sales Service Department.

6.1 Troubleshooting guide

The troubleshooting guide is specially designed for assisting user in finding and eliminating faults occurring during operation of the instrument. In order to accurately and quickly find and eliminate faults, please first carefully read this manual to know normal operation and preventive maintenance of the instrument.

Generally, troubleshooting should follow the three steps below:

Step 1: Confirm the fault

The operator should identify the cause of the fault before trying to eliminate it.

Step 2: Classify the fault

Basically, faults of the instrument can be classified into the following three types:

1. Faults related to hardware.
2. Faults related to software.
3. Test failures related to sample analysis.

In case of faults related to hardware and software, the instrument can be repaired only by qualified engineers of or authorized by AGAPPE. Test failures related to sample analysis can be eliminated by the operator under the guidance of AGAPPE’s engineer.

Step 3: Eliminate the fault

Our service engineer will take proper measures to eliminate the fault. If the operator can eliminate the fault by himself/herself under the help of AGAPPE’s engineer, there will be much less time delayed.

6.2 Seek technical support

If AGAPPE's technical support is needed when the instrument malfunctions, please call our After-sales Service Department; see Copyright and Statement for the phone number and fax. When seeking technical support, please provide a detailed and clear description of the problem and relevant information as follows:

1. Model of the instrument.
2. SN on the instrument.
3. A detailed description of the fault symptom and operating environment (for example, what operation is performed under which window and in what status).
4. Data and report related to the fault.

6.3 Faults of PC operating system and hardware

No.	Symptom	Possible Cause	Solution
1	The PC cannot be started.	1) Power supply is not connected.	1) Connect the power cord.
2	The PC cannot enter the system.	1) Memory failure. 2) Hard disk drive failure.	1) Replace the memory. 2) Detect whether any bad track exists. If yes, first repair the bad track; if it is unrepairable, please replace the hard disk drive, and copy data to the new hard disk drive.
3	PC crash	1) Hard disk drive failure. 2) USB port failure. 3) Virus infection.	1) See the solution for No. 2 fault in this table. 2) Disconnect the plug-and-play device from the USB port, and restart the PC. 3) Upgrade the antivirus software to kill virus; if the problem remains, please reinstall or restore

			the operating system.
4	The PC runs slowly.	1) Virus infection. 2) Camera disconnection.	1) Upgrade the antivirus software to kill virus; if the problem remains, please reinstall or restore the operating system. 2) Check the connection.

6.4 Faults of applied parts of the instrument

No.	Fault	Possible Cause	Solution
1	Power-on failure	The fuse is damaged.	Replace the fuse. If the problem remains, please contact the After-sales Service Department of Agappe
2	Absence of Cleaning solution	Cleaning solution is used up or the liquid level sensor is damaged.	Replenish the Cleaning solution bottle with Cleaning solution. If the problem remains, please contact the After-sales Service Department of Agappe
3	Absence of test strip	Test strips are used up or the strip selection mechanism is damaged.	Add test strips to the test strip compartment. If the problem remains, please contact the After-sales Service Department of Agappe
4	Printing failure	The printer is damaged or software settings are	Correct the software settings or load the printing paper.

		improper. Absence of printing paper.	If the problem remains, please contact the After-sales Service Department of Agappe
5	Black screen	The display screen is damaged or power failure occurs.	Replace the fuse. If the problem remains, please contact the After-sales Service Department of Agappe
6	Motor getting stuck	The motor or the limit optocoupler is damaged.	Restart the instrument. If the problem remains, please contact the After-sales Service Department of Agappe
7	Image fuzziness	The focal length of the microscope is not properly adjusted.	Adjust the focal length.
8	Reagent alarm	There is no reagent in the pipe.	Change the reagent and perform priming.
9	Camera initialization failure.	Failure of camera data cable. Failure of camera drive.	Replace the data cable. Reinstall the camera driver.
10	Motor not reset upon startup.	Failure of motor reset sensor.	Replace the sensor.
11	After sampling, the sample in the counting cell cannot	Failure of solenoid valve at both ends of the counting cell.	Replace the solenoid valve.

	become still after a long period of time.		

In any of the following situations, you can contact our After-sales Service Department, and we will offer solutions as soon as possible.

- The problem cannot be solved with the solutions listed in the table above;
- The problem is caused by use of damaged accessories;
- The problem cannot be solved since it is obviously attributable to the instrument itself;
- Main components of the instrument are damaged, etc.

Caution: When replacing the fuse, please first turn off the instrument and then disconnect the power cord.

Chapter 7 Prevention, restriction and hazards

Improper use of any instrument will result in failure to achieve the expected result or even cause injury of the operator or other persons. Therefore, before operation and use of the instrument, it is necessary to standardize the rules for use and perfect the conditions for use to prevent the occurrence of hazards and achieve the optimum operation effect of the instrument.

7.1 Restriction on use

a) Automatic Urine Analysis System is only used for analysis to provide reference for in vitro diagnosis.

b) Any person using, moving, installing or maintaining the instrument must carefully read this manual and strictly follow the requirements specified in the manual. Otherwise, occurrence of accidents could be caused, and our free warranty will be voided.

c) Repair of the instrument must be permitted by AGAPPE. For replacement of any part, the part specified by AGAPPE should be used; otherwise, occurrence of accidents could be caused, and our free warranty will be voided.

d) Failure to carry out periodic maintenance according to the maintenance procedure described in Chapter 5 Maintenance of Instrument of this manual will affect the measurement and analysis results and shorten the service life of the instrument, and could result in occurrence of accidents. Also, our free warranty will be voided.

⚠ WARNING

- Parts should be checked or provided only by AGAPPE or its authorized agency.

7.2 Restriction on installation

a) Initial installation must be done by qualified engineers with AGAPPE's authorization.

b) The instrument must be placed on a horizontal and steady workbench without exposure to direct sunlight; it should be kept away from air vents that are too cold/hot, and also away from equipment like dryers, centrifuges, X-ray equipment, copiers and ultrasonic cleaning machines.

c) The reagent should be placed at the same height as the instrument, or kept on the same plane as the instrument.

d) To maintain a good ventilation, certain space must be reserved around the analyzer. There should be a distance of at least 40 cm between the instrument and surrounding objects to facilitate ventilation. The installation surface should be flat without vibration and have an area of at least 1500 mm×800 mm. There should be adequate space around the instrument to facilitate repair and maintenance.

e) Before initial operation of the instrument, please confirm the following: whether the pipeline of each reagent bottle is properly and reliably connected; whether the externally connected pipe is bent, affecting smooth liquid flow; whether liquid waste is discharged to a proper container.

f) When the instrument is in power-on state, it is forbidden to unplug/plug any electrical connector of the instrument. To prevent electricity interruption and protect user's personal safety, please confirm that the earthing conductor of the instrument housing is well grounded.

⚠ WARNING

- Any person not authorized or qualified to repair the instrument shall not loosen the screws in the instrument housing or open the door lock at any time; otherwise, user himself/herself shall be responsible for any problem arising therefrom!

7.3 Personal protection and infection control

a) During sample test and routine maintenance, user must observe the laboratory requirements or the operation requirements for clinical test, and wear medical gloves and safety goggles to avoid direct contact with samples.

b) All clinical samples, QCs and calibrators could contain human urine and are potentially infectious. Therefore, when handling these materials, user must observe the established laboratory or clinical operation procedure and wear work clothes, medical gloves and safety goggles. Do not smoke, eat or drink in the work area; do not blow or suck the working pipe with your mouth.

c) Since urine samples and liquid waste produced by the instrument contain potential contaminants and may have biological and chemical hazards, user shall take extreme care during handling. Please observe the applicable regulations of the local government during cleaning, treatment and discharge.

d) Do not pour the remaining unused reagent to a new reagent tank (bottle) to prevent contamination of the new reagent.

e) Reagents, calibrators and QCs should be stored and kept according to the instructions for use. User of the instrument is obliged to establish and maintain effective storage and requisitioning measures to prevent use of reagents, calibrators and QCs beyond their shelf-life as well as deterioration, misuse or eating by mistake of these materials. Reagents should not be stored in an environment that is too cold/hot.

7.4 Names and contents of toxic or hazardous substances in the product

Component	Hazardous Substances					
	Pb	Hg	Cd	Cr^{6+}	PBB	PBDE
PCBA	○	○	○	○	○	○
Plastic	○	○	○	○	○	○

Metal	○	○	○	○	○	○

○: Indicates that the content of this toxic or hazardous substance in all homogeneous materials of the corresponding part meets the limit specified in SJ/T 11363-2006.

×: Indicates that the content of this toxic or hazardous substance in a homogeneous material of the corresponding part is above the limit specified in SJ/T 11363-2006.

⚠ NOTE

- If the storage temperature is below 0℃, the reagent will be frozen. Once the reagent is frozen, stop using it.
- Reagents should not be exposed to direct sunlight to prevent reagent evaporation and contamination. Tightly close the cover of reagent container; when the pipe passes through the reagent cover, the hole diameter on the reagent cover should be minimized to avoid the occurrence of evaporation and contamination.

Appendix 1: List of main components

No.	Name & Specification	Total quantity	Unit
1	Main unit of Automatic Urine Analysis System	1	Set
	Monitor	1	Set
	Keyboard and mouse	1	Set
2	User Manual	1	Set
3	Certificate of Conformity/Warranty Card	1	Sheet
4	KU-2800 Urine Analyzer Operating Procedure	2	Sheet
5	KU-2800 Urine Analyzer Maintain Procedure	2	Sheet
6	Power cord	2	Set
7	Urine Reagent Strips	1	Bottle
8	5L liquid waste tank	1	PCS
9	Liquid waste alarm unit	1	Set
10	Waste strip box	1	Set
11	Test tube rack	10	PCS
12	Sample Diluent pipe	1	Set
13	Cleaning Solution pipe	1	Set
14	Liquid waste pipe	1	Set
15	Urine Sample Collection Tube	1	Bag
16	Toolkit	1	Set

Note: Subject to the packing list provided with the instrument.

Appendix 2: Color identification list

No.	English Name
1	Red
2	Yellow
3	Orange
4	Green
5	Transparent
6	Unknow

Appendix 3: Test item list

No.	Name	Abbreviation
1	Leukocytes	LEU
2	Nitrite	NIT
3	Urobilinogen	URO
4	Bilirubin	BIL
5	Vitamin C	VC
6	Protein	PRO
7	Occult Blood	BLD
8	pH	pH
9	Specific gravity	SG
10	Glucose	GLU
11	Ketone Body	KET
12	Microalbumin	MA
13	Creatinine	Cr
14	Calcium	Ca

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MISPA U280-User Manual-R0.1 Company reserves the right to change any design and technical features of the product at any time, If needed.